Free Standing Enclosure (Vault file-version 991056) native snapshot: parts inventory

Scene `free-standing-enclosure-991056`: Free Standing Enclosure (Vault file-version 991056) native snapshot. Document `free-standing-enclosure-991056-inventory`. Generated 2026-10-22T18:37:42Z.

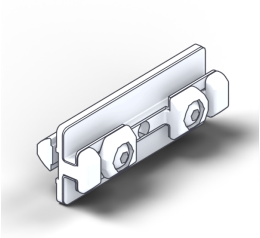
DRAFT

Parts inventory and overview only: no assembly sequence has been authored or reviewed for this scene.
Geometry and hierarchy come from the CAD export named in the build record. Quantities are CAD instance counts, not purchase quantities, kits, stock or an approved bill of materials.
Project note: The tank(s) and the lower light box are to be replaced in the modified build by two Alienware AW2725Q monitors under an acrylic panel and bowl frames (Peter, 2026-10-22; intent, not a designer confirmation; the light-box occurrences are not yet identified). CAD counts are unchanged.
Subassembly pages and joint close-ups are inspection views of parts at their CAD positions; they are not assembly steps and imply no sequence, fasteners or tool access.
Supplementary items (20 rows, 16 for the existing design, 4 for new design in the appendix only) are hardware not modelled in CAD: listed separately for review, never added to CAD quantities, unknown quantities left unknown; not an approved bill of materials.
CAD ambiguity: 1 pair(s) of occurrences of the same part share one position (SHULZE 2W.353 TANK). All source instances are kept and counted; review the intended configuration before treating the count as stock.
Stock is unknown until counted and recorded (on_hand_each with counted_by and counted_date). The counted box is for a paper count only.

Parts

CAD qty is the number of instances of the part in the CAD model (reference-only geometry excluded). It is not a purchase quantity: purchased kits, pack sizes, spares and stock are not reconciled. Write the quantity you count in the box.

1 MCMASTER 5537T293 T-Slotted Framing

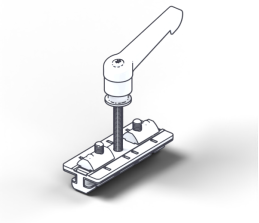


overall 80 × 32 × 28.4 mm
Zinc · 9 bodies
on hand unknown
intended configuration confirmed: no

2 CAD qty 2 each

counted

2 MCMASTER 5537T429 T-Slotted Framing



overall 103 × 99.2 × 32 mm
Zinc · 11 bodies
on hand unknown
intended configuration confirmed: no

2 CAD qty 2 each

counted

3 SLLB Enclosure 15in 47065T808 T-Slotted Framing



overall 381 × 38.1 × 38.1 mm
ALUMINUM, 6061-T651
on hand unknown
intended configuration confirmed: no

2 CAD qty 2 each

counted

4 SLLB Enclosure 22in 47065T808 T-Slotted Framing



overall 559 × 38.1 × 38.1 mm
ALUMINUM, 6061-T651
on hand unknown
intended configuration confirmed: no

2 CAD qty 2 each

counted

5 SLLB Enclosure 41in 47065T808 T-Slotted Framing

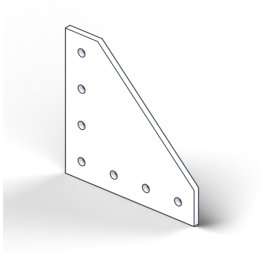


overall 1032 × 38.1 × 38.1 mm
ALUMINUM, 6061-T651
on hand unknown
intended configuration confirmed: no

2 CAD qty 2 each

counted

6 MCMASTER 3136N149 T-Slotted Framing

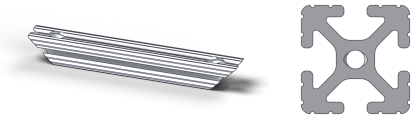


overall 152 × 152 × 6.3 mm
ALUMINUM, 6061-T651
on hand unknown
intended configuration confirmed: no

2 CAD qty 2 each

counted

7 MCMASTER 47065T27 T-Slotted Framing

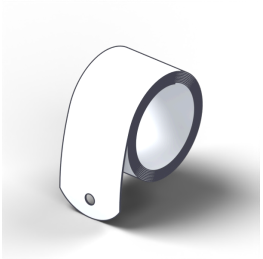


overall 305 × 38.1 × 38.1 mm
ALUMINUM, 6061-T651
on hand unknown
intended configuration confirmed: no

4 CAD qty 4 each

counted

8 MCMASTER 9293K121 Constant-Force Spring

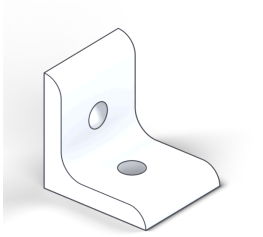


overall 59.9 × 52.6 × 31.8 mm
Generic
on hand unknown
intended configuration confirmed: no

2 CAD qty 2 each

counted

9 MCMASTER 47065T845 Silver Corner Bracket

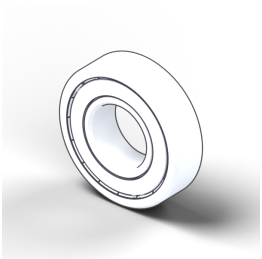


overall 38.1 × 38.1 × 33.3 mm
ALUMINUM, 6061-T651
on hand unknown
intended configuration confirmed: no

4 CAD qty 4 each

counted

10 MCMASTER 5972K164 Ball Bearing



overall 22 × 22 × 6 mm
P/N MCMASTER 5972K164_Ball Bearing · Generic · 14 bodies
on hand unknown
intended configuration confirmed: no

4 CAD qty 4 each

counted

11 MCMASTER 90265A164 18-8 Stainless Steel Shoulder Screw

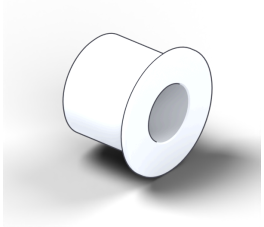


overall 45.3 × 16 × 16 mm
P/N MCMASTER 90265A164_18-8 Stainless Steel Shoulder Screw · Generic
on hand unknown
intended configuration confirmed: no

2 CAD qty 2 each

counted

12 SPRING ROLLER

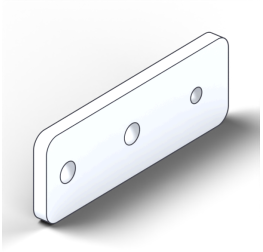


overall 48 × 48 × 34.3 mm (1.89 × 1.89 × 1.35 in)
ABS, BLACK
on hand unknown
intended configuration confirmed: no

2 CAD qty 2 each

counted

13 ROLLER MOUNT PLATE

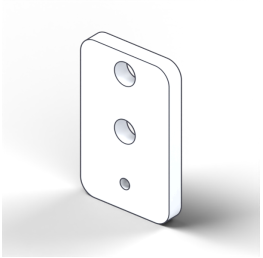


overall 108 × 38.1 × 6.3 mm (4.25 × 1.5 × 0.25 in)
ALUMINUM, 6061-T651
on hand unknown
intended configuration confirmed: no

2 CAD qty 2 each

counted

14 SPRING MOUNT PLATE

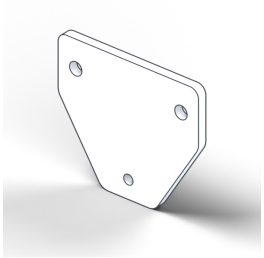


overall 60 × 38.1 × 6.3 mm (2.362 × 1.5 × 0.25 in)
P/N SPRING MOUNT PLATE · ALUMINUM, 6061-T651
on hand unknown
intended configuration confirmed: no

2 CAD qty 2 each

counted

15 CAMERA MOUNT PLATE

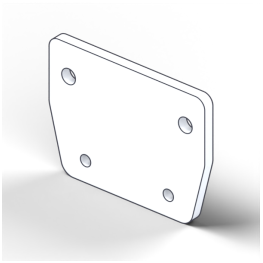


overall 102 × 88.9 × 6.3 mm (4 × 3.5 × 0.25 in)
P/N CAMERA MOUNT PLATE · ALUMINUM, 6061-T651
on hand unknown
intended configuration confirmed: no

1 CAD qty 1 each

counted

16 CAMERA MOUNT PLATE - TELEDYNE KINETIX



overall 102 × 76.2 × 6.3 mm (4 × 3 × 0.25 in)
ALUMINUM, 6061-T651
on hand unknown
intended configuration confirmed: no

1 CAD qty 1 each

counted

17 TSLOT EXTRUSION 1X40.5 copy

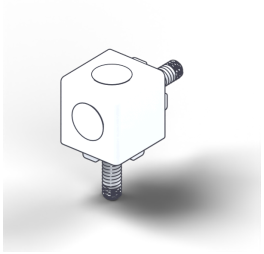


overall 1029 × 25.4 × 25.4 mm (40.5 × 1 × 1 in)
Generic
on hand unknown
intended configuration confirmed: no

8 CAD qty 8 each

counted

18 T SLOT CORNER 47065T242

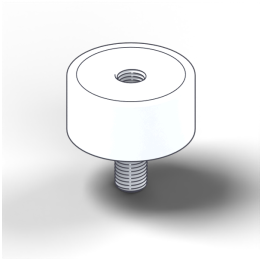


overall 47.4 × 47.4 × 25.2 mm (1.868 × 1.868 × 0.992 in)
P/N 47065T242_T-Slotted Framing · Generic · 5 bodies
on hand unknown

8 CAD qty 8 each

counted

19 MCMASTER 9378K24 Natural Rubber Vibration-Damping Sandwich

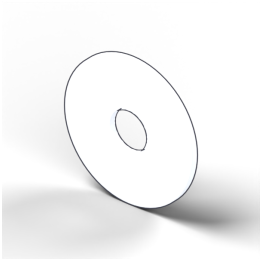


overall 25.7 × 25.4 × 25.4 mm (1.013 × 1 × 1 in)
P/N 9378K24_Natural Rubber Vibration-Damping Sandwich Mount · Generic · 3 bodies
on hand unknown
intended configuration confirmed: no

8 CAD qty 8 each

counted

20 MCMASTER 90313A107 18-8 Stainless Steel Oversized Washer

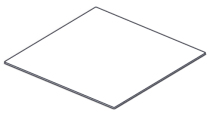


overall 25.4 × 25.4 × 1.3 mm (1 × 1 × 0.05 in)
P/N 90313A107_18-8 Stainless Steel Oversized Washer · STAINLESS STEEL, 18-8
on hand unknown
intended configuration confirmed: no

8 CAD qty 8 each

counted

21 TANK SUPPORT PLATFORM

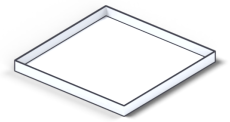


overall 1080 × 1080 × 12.7 mm (42.5 × 42.5 × 0.5 in)
P/N TANK SUPPORT PLATFORM · ACRYLIC, CLEAR
on hand unknown
intended configuration confirmed: no

2 CAD qty 2 each

counted

22 SHULZE 2W.353 TANK



overall 991 × 991 × 82.6 mm (39 × 39 × 3.25 in)
Acrylic, Clear
on hand unknown
intended configuration confirmed: no
⚠ 1 coincident instance pair(s) in CAD: review

3 CAD qty 3 each

counted

23 TANK SUPPORT BRACKET



overall 114 × 63.5 × 6.3 mm (4.5 × 2.5 × 0.25 in)
ALUMINUM, 6061-T651
on hand unknown
intended configuration confirmed: no

8 CAD qty 8 each

counted

24 SLLB Enclosure 44in 47065T808 T-Slotted Framing

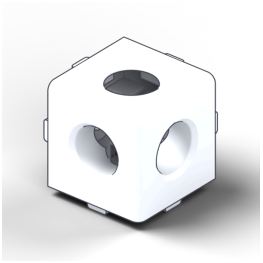


overall 1118 × 38.1 × 38.1 mm
P/N SLLB Enclosure 44in 47065T808_T-Slotted Framing · Generic
on hand unknown
intended configuration confirmed: no

11 CAD qty 11 each

counted

25 MCMASTER 47065T245



overall 41.1 × 41.1 × 41.1 mm
P/N MCMASTER 47065T245 · Generic · 4 bodies
on hand unknown
intended configuration confirmed: no

8 CAD qty 8 each

counted

26 SLLB Enclosure 96in 47065T808 T-Slotted Framing

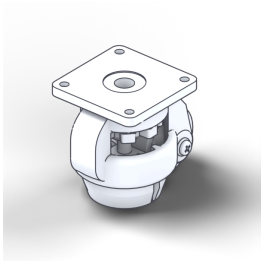


overall 2438 × 38.1 × 38.1 mm
Generic
on hand unknown
intended configuration confirmed: no

4 CAD qty 4 each

counted

27 HOLKIE LEVELING CASTER



overall 104 × 86.2 × 73.5 mm (4.094 × 3.395 × 2.893 in)
P/N HOLKIE LEVELING CASTER · Generic · 10 bodies
on hand unknown

4 CAD qty 4 each

counted

28 CASTER MOUNTING PLATE

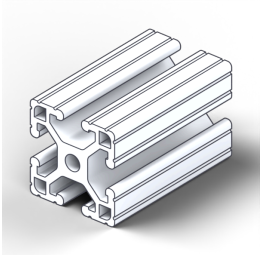


overall 100 × 100 × 6.3 mm (3.953 × 3.953 × 0.25 in)
ALUMINUM, 6061-T651
on hand unknown
intended configuration confirmed: no

4 CAD qty 4 each

counted

29 SLLB Enclosure 2.5in 47065T808 T-Slotted Framing

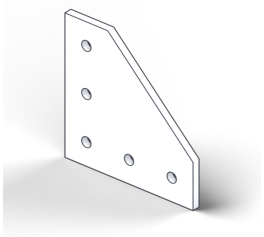


overall 63.5 × 38.1 × 38.1 mm
Generic
on hand unknown
intended configuration confirmed: no

2 CAD qty 2 each

counted

30 47065T271 ALUMINUM T-SLOTTED FRAMING EXTRUSION

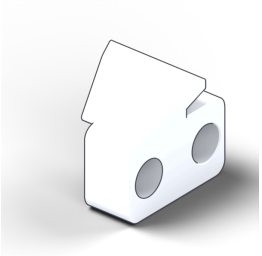


overall 114 × 114 × 6.3 mm
Aluminum
on hand unknown
intended configuration confirmed: no

8 CAD qty 8 each

counted

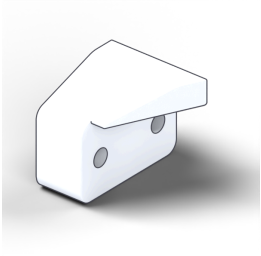
31 PANEL LOWER SUPPORT LEFT



overall 63.5 × 52.7 × 51.1 mm (2.5 × 2.075 × 2.01 in)
Generic
on hand unknown
intended configuration confirmed: no

4 CAD qty 4 each counted

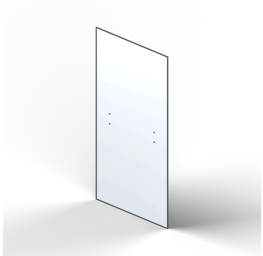
32 PANEL LOWER SUPPORT RIGHT



overall 63.5 × 52.7 × 51.1 mm (2.5 × 2.075 × 2.01 in)
ABS, BLACK
on hand unknown
intended configuration confirmed: no

4 CAD qty 4 each counted

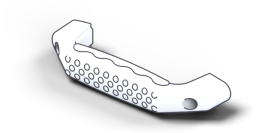
33 ENCLOSURE PANEL



overall 2438 × 1143 × 3.2 mm (96 × 45 × 0.125 in)
ALUMINUM COMPOSITE MATERIAL
on hand unknown
intended configuration confirmed: no

4 CAD qty 4 each counted

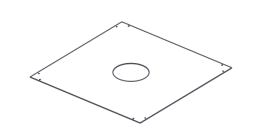
34 MCMASTER 1950A67 Unthreaded-Hole Pull Handle



overall 170 × 51.1 × 31.2 mm (6.71 × 2.014 × 1.23 in)
P/N MCMASTER_1950A67_Unthreaded-Hole Pull Handle · Generic
on hand unknown
intended configuration confirmed: no

8 CAD qty 8 each counted

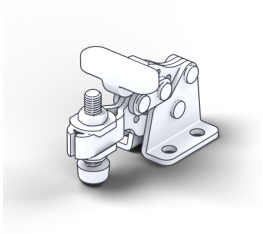
35 ENCLOSURE TOP PANEL



overall 1194 × 1194 × 3.2 mm (47 × 47 × 0.125 in)
P/N ENCLOSURE TOP PANEL · ALUMINUM COMPOSITE MATERIAL
on hand unknown
intended configuration confirmed: no

1 CAD qty 1 each counted

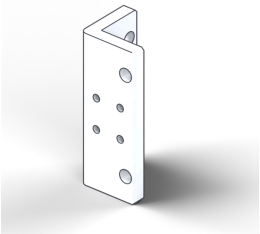
36 MCMASTER 5004A11 Compact Hold-Down Toggle Clamp



overall 56.5 × 35.7 × 28.4 mm (2.224 × 1.407 × 1.12 in)
P/N 5004A11_Compact Hold-Down Toggle Clamp · Generic · 20 bodies
on hand unknown

8 CAD qty 8 each counted

37 TOGGLE CLAMP MOUNT



overall 76.2 × 30.5 × 30.5 mm (3 × 1.2 × 1.2 in)
ABS, BLACK
on hand unknown
intended configuration confirmed: no

4 CAD qty 4 each counted

The counted box is for a paper count only; recorded stock is on_hand_each in the project's manifests.

Supplementary items (not modelled in CAD)

Supplementary items are hardware, consumables or purchased items that are not modelled in CAD. They are listed for review and are never added to CAD quantities; an unknown quantity stays unknown. This is not an approved bill of materials. Status: draft: rows converted from the project procurement sheet; candidate item families, not an approved BOM; quantities unknown unless stated; never added to CAD counts.

#	Item	Kind, status	Required	Stock	Included with	Config. confirmed	counted
S1	**Caster-to-mounting-plate fastening** `SUP-001` joint `caster_to_plate` attaches to: CASTER ASSEMBLY:1, CASTER ASSEMBLY:2, CASTER ASSEMBLY:3, CASTER ASSEMBLY:4	item family, no SKU candidate, requires review	required unknown (family, not specified); basis: 4 caster assemblies; attachment points per caster unverified	on hand unknown	—	no	☐
S2	**Caster-plate-to-frame fastening** `SUP-002` joint `caster_plate_to_frame` attaches to: FREE STANDING FRAME:1, CASTER ASSEMBLY:1, CASTER ASSEMBLY:2, CASTER ASSEMBLY:3, CASTER ASSEMBLY:4	item family, no SKU candidate, requires review	required unknown (family, not specified); basis: 4 plates; frame attachments per plate unverified	on hand unknown	—	no	☐
S3	**Pull-handle-to-panel fastening** `SUP-003` joint `handle_to_panel` attaches to: PANEL ASSEMBLY:1, PANEL ASSEMBLY:2, PANEL ASSEMBLY:3, PANEL ASSEMBLY:4	item family, no SKU candidate, requires review	required unknown (family, not specified); basis: 8 handles on 4 panels; fastening points per handle unverified	on hand unknown	—	no	☐
S4	**Toggle-clamp-to-mount fastening** `SUP-004` joint `clamp_to_mount` attaches to: TOGGLE CLAMP ASSEMBLY:1, TOGGLE CLAMP ASSEMBLY:2, TOGGLE CLAMP ASSEMBLY:3, TOGGLE CLAMP ASSEMBLY:4	item family, no SKU candidate, requires review	required unknown (family, not specified); basis: 8 clamps on 4 mounts; points per clamp unverified	on hand unknown	—	no	☐
S5	**Toggle-clamp-mount-to-frame fastening** `SUP-005` joint `clamp_mount_to_frame` attaches to: TOGGLE CLAMP ASSEMBLY:1, TOGGLE CLAMP ASSEMBLY:2, TOGGLE CLAMP ASSEMBLY:3, TOGGLE CLAMP ASSEMBLY:4	item family, no SKU candidate, requires review	required unknown (family, not specified); basis: 4 mounts; points per mount unverified	on hand unknown	—	no	☐
S6	**Panel-lower-support fastening** `SUP-006` joint `lower_support_to_frame` attaches to: FREE STANDING FRAME:1	item family, no SKU candidate, requires review	required unknown (family, not specified); basis: 4 left + 4 right supports; points per support unverified	on hand unknown	—	no	☐
S7	**Main-frame connector hardware and joint completeness** `SUP-007` joint `main_frame_joints` attaches to: FREE STANDING FRAME:1	item family, no SKU candidate, requires review	required unknown (family, not specified); basis: 11 long rails + 4 columns + 2 short rails; joint count requires review	on hand unknown	—	no	☐
S8	**Camera-frame connector and bracket hardware**	item family, no SKU	required unknown (family,	on hand unknown	—	no	☐

#	Item	Kind, status	Required	Stock	Included with	Config. confirmed	counted
	SUP-008 joint camera_frame_joints attaches to: CAMERA FRAME:1	candidate, requires review	not specified); basis: 4 named 47065T845 brackets plus other connector/ sliding parts				
S9	**Tank-support-frame corner hardware** SUP-009 joint support_frame_corners attaches to: TANK SUPPORT FRAME:1, TANK SUPPORT FRAME:2	item family, no SKU candidate, requires review	required unknown (family, not specified); basis: 8 modeled corner connectors across 2 support frames	on hand unknown	—	no	☐
S10	**Tank-support-bracket attachment hardware** SUP-010 joint support_brackets attaches to: TANK SUPPORT FRAME:1, TANK SUPPORT FRAME:2	item family, no SKU candidate, requires review	required unknown (family, not specified); basis: 8 brackets across 2 support frames	on hand unknown	—	no	☐
S11	**Vibration-mount retention hardware** SUP-011 joint vibration_mounts attaches to: TANK SUPPORT FRAME:1, TANK SUPPORT FRAME:2	item family, no SKU candidate, requires review	required unknown (family, not specified); basis: 8 mounts + 8 oversized washers modeled	on hand unknown	—	no	☐
S12	**Camera, spring and roller plate attachment hardware** SUP-012 joint camera_plates attaches to: CAMERA FRAME:1	item family, no SKU candidate, requires review	required unknown (family, not specified); basis: 2 roller plates + 2 spring plates + 2 different camera plates	on hand unknown	—	no	☐
S13	**Spring end and roller/bearing retention hardware** SUP-013 joint spring_roller attaches to: CAMERA FRAME:1	item family, no SKU candidate, requires review	required unknown (family, not specified); basis: 2 springs + 2 rollers + 4 bearings + 2 shoulder screws modeled	on hand unknown	—	no	☐
S14	**Camera-to-mount attachment** SUP-014 joint camera_attachment attaches to: CAMERA FRAME:1	item family, no SKU candidate, requires review	required unknown (family, not specified); basis: Camera plate variants present; actual camera selection unconfirmed	on hand unknown	—	no	☐
S15	**Top-panel retention and edge/ support interface** SUP-015 joint top_panel attaches to: ENCLOSURE TOP PANEL:1	item family, no SKU candidate, requires review	required unknown (family, not specified); basis: 1 enclosure top panel	on hand unknown	—	no	☐
S16	**Removable-panel retention interface** SUP-016 joint panel_retention attaches to: FREE STANDING FRAME:1, PANEL ASSEMBLY:1, PANEL ASSEMBLY:2, PANEL ASSEMBLY:3, PANEL ASSEMBLY:4, TOGGLE CLAMP ASSEMBLY:1, TOGGLE CLAMP	item family, no SKU candidate, requires review	required unknown (family, not specified); basis: 4 panels, 8 lower supports, 8 toggle clamps	on hand unknown	—	no	☐

#	Item	Kind, status	Required	Stock	Included with	Config. confirmed	counted
	ASSEMBLY:2, TOGGLE CLAMP ASSEMBLY:3, TOGGLE CLAMP ASSEMBLY:4						

4 new-design row(s) are in the appendix (not printed in this compact document).

Subassembly 1: CASTER ASSEMBLY:1

`FREE STANDING FRAME:1/CASTER ASSEMBLY:1` under `FREE STANDING FRAME:1`

inspection view: occurrences at their CAD poses; not an assembly step; no sequence, fasteners or tool access implied.

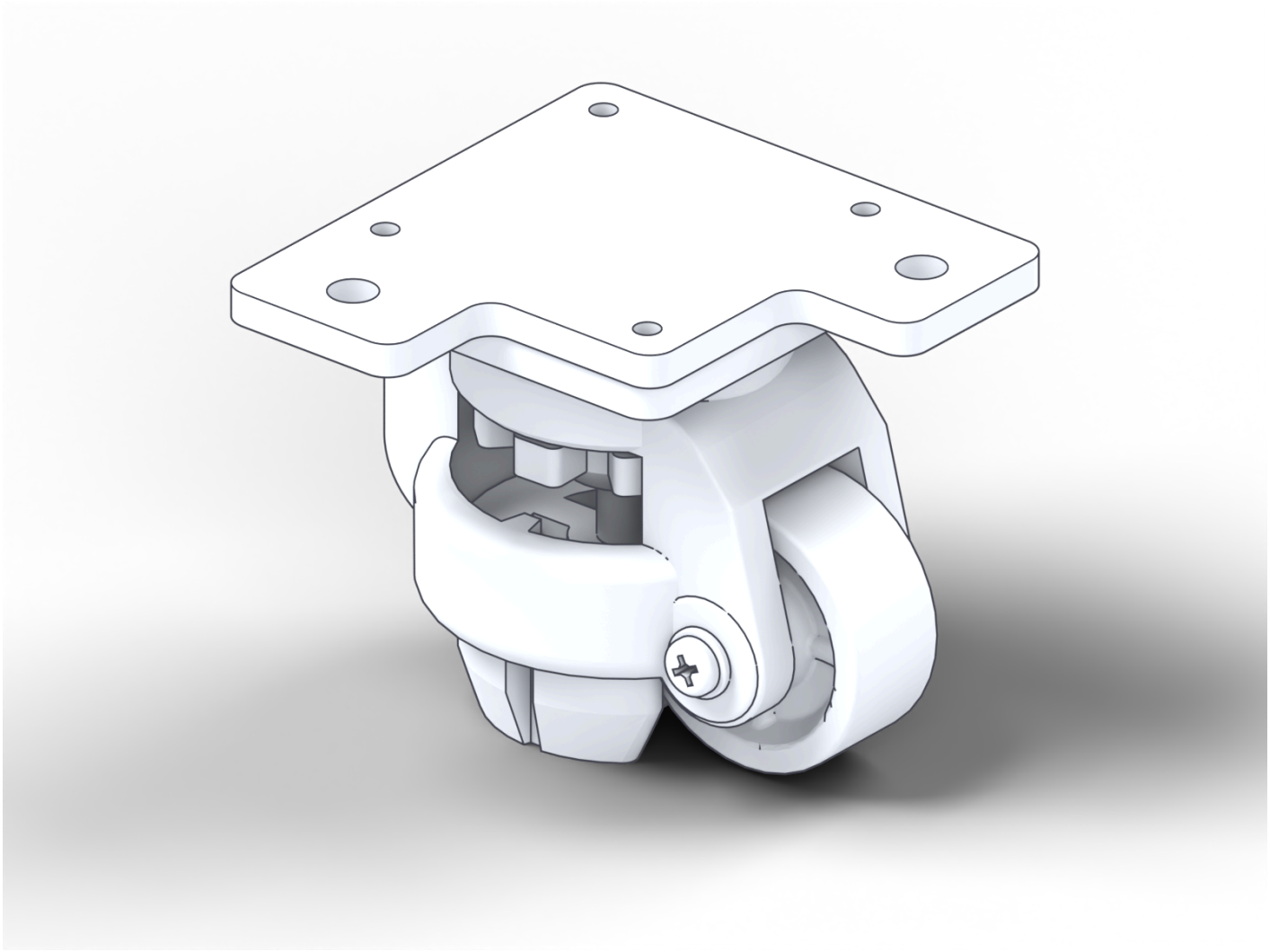


Figure 1: Subassembly 1: CASTER ASSEMBLY:1, parts at their CAD positions (inspection view).

#	Picture	Part	In this subassembly
27		HOLKIE LEVELING CASTER	×1
28		CASTER MOUNTING PLATE	×1

2 occurrence(s); ids in manual.json.

Hardware to review at this joint (not an approved BOM)

- S1 Caster-to-mounting-plate fastening · candidate, requires review · required unknown (family, not specified); basis: 4 caster assemblies; attachment points per caster unverified · joint `caster_to_plate`
- S2 Caster-plate-to-frame fastening · candidate, requires review · required unknown (family, not specified); basis: 4 plates; frame attachments per plate unverified · joint `caster_plate_to_frame`

Subassembly 2: PANEL ASSEMBLY:1

`PANEL ASSEMBLY:1` under `root`

inspection view: occurrences at their CAD poses; not an assembly step; no sequence, fasteners or tool access implied.



Figure 2: Subassembly 2: PANEL ASSEMBLY:1, parts at their CAD positions (inspection view).

#	Picture	Part	In this subassembly
33		ENCLOSURE PANEL	×1
34		MCMASTER 1950A67 Unthreaded-Hole Pull Handle	×2

3 occurrence(s); ids in manual.json.

Hardware to review at this joint (not an approved BOM)

- S3 Pull-handle-to-panel fastening · candidate, requires review · required unknown (family, not specified); basis: 8 handles on 4 panels; fastening points per handle unverified · joint `handle_to_panel`
- S16 Removable-panel retention interface · candidate, requires review · required unknown (family, not specified); basis: 4 panels, 8 lower supports, 8 toggle clamps · joint `panel_retention`

Subassembly 3: TOGGLE CLAMP ASSEMBLY:1

`TOGGLE CLAMP ASSEMBLY:1` under `root`

inspection view: occurrences at their CAD poses; not an assembly step; no sequence, fasteners or tool access implied.

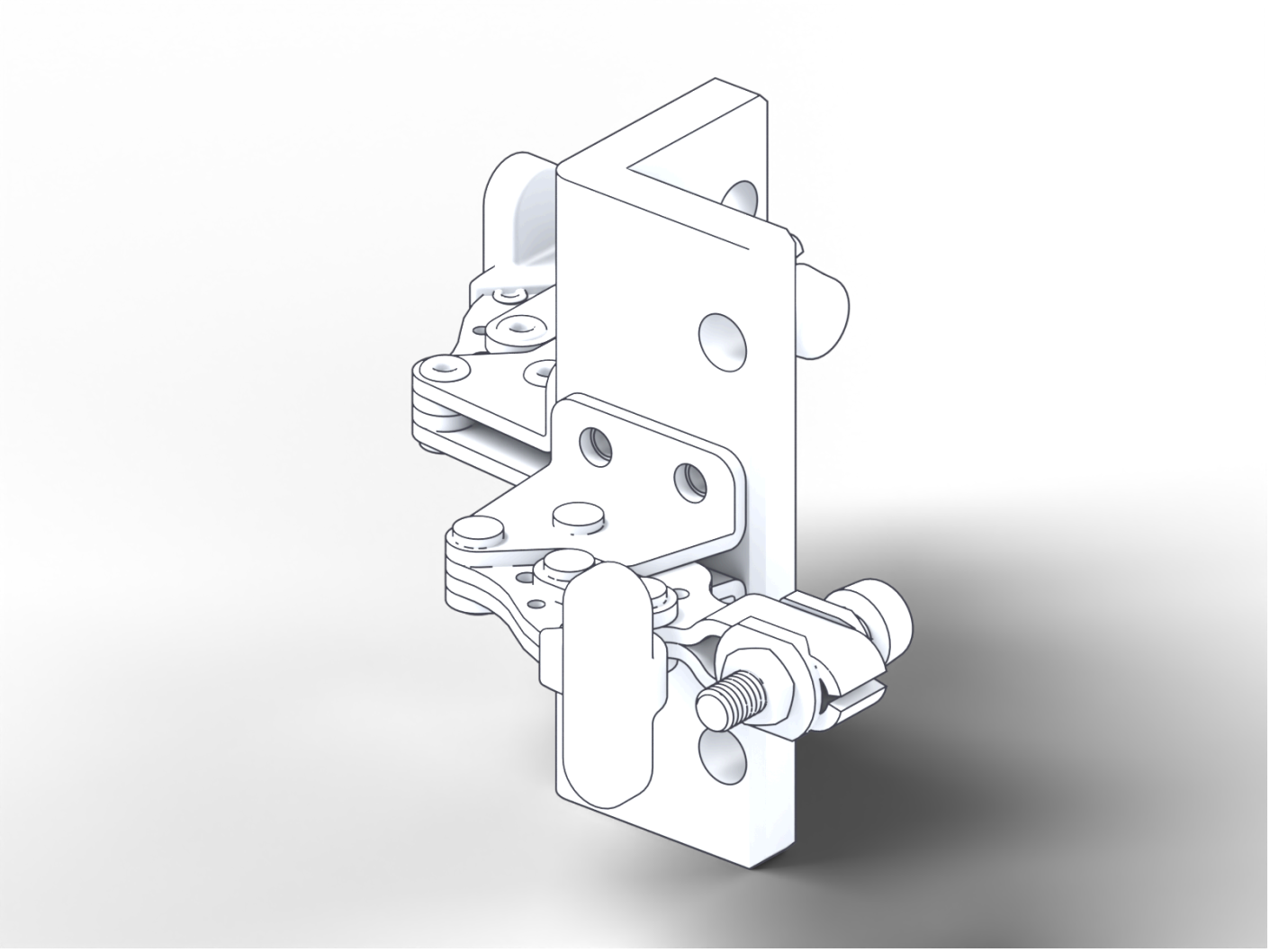
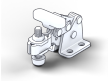


Figure 3: Subassembly 3: TOGGLE CLAMP ASSEMBLY:1, parts at their CAD positions (inspection view).

#	Picture	Part	In this subassembly
36		MCMASTER 5004A11 Compact Hold-Down Toggle Clamp	×2
37		TOGGLE CLAMP MOUNT	×1

3 occurrence(s); ids in manual.json.

Hardware to review at this joint (not an approved BOM)

- S4 Toggle-clamp-to-mount fastening · candidate, requires review · required unknown (family, not specified); basis: 8 clamps on 4 mounts; points per clamp unverified · joint `clamp_to_mount`
- S5 Toggle-clamp-mount-to-frame fastening · candidate, requires review · required unknown (family, not specified); basis: 4 mounts; points per mount unverified · joint `clamp_mount_to_frame`
- S16 Removable-panel retention interface · candidate, requires review · required unknown (family, not specified); basis: 4 panels, 8 lower supports, 8 toggle clamps · joint `panel_retention`

Subassembly 4: TANK SUPPORT FRAME:1

`TANK SUPPORT FRAME:1` under `root`

inspection view: occurrences at their CAD poses; not an assembly step; no sequence, fasteners or tool access implied.

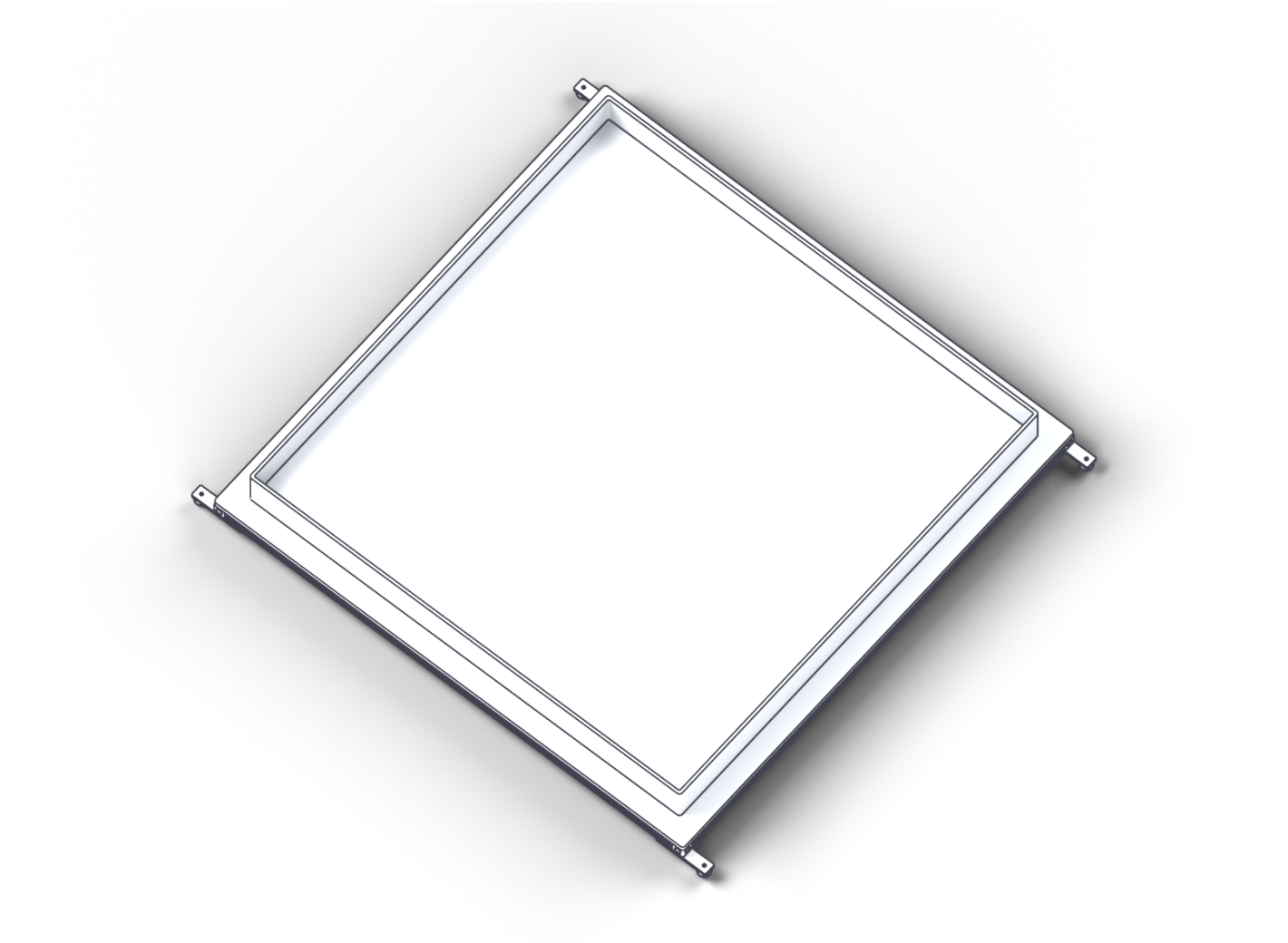
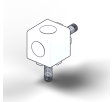
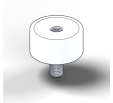


Figure 4: Subassembly 4: TANK SUPPORT FRAME:1, parts at their CAD positions (inspection view).

#	Picture	Part	In this subassembly
17		TSLOT EXTRUSION 1X40.5 copy	×4
18		T SLOT CORNER 47065T242	×4
19		MCMASTER 9378K24 Natural Rubber Vibration-Damping Sandwich Mount	×4
20		MCMASTER 90313A107 18-8 Stainless Steel Oversized Washer	×4
21		TANK SUPPORT PLATFORM	×1

#	Picture	Part	In this subassembly
22		SHULZE 2W.353 TANK	×1
23		TANK SUPPORT BRACKET	×4

22 occurrence(s); ids in manual.json.

Hardware to review at this joint (not an approved BOM)

- S9 Tank-support-frame corner hardware · candidate, requires review · required unknown (family, not specified); basis: 8 modeled corner connectors across 2 support frames · joint `support_frame_corners`
- S10 Tank-support-bracket attachment hardware · candidate, requires review · required unknown (family, not specified); basis: 8 brackets across 2 support frames · joint `support_brackets`
- S11 Vibration-mount retention hardware · candidate, requires review · required unknown (family, not specified); basis: 8 mounts + 8 oversized washers modeled · joint `vibration_mounts`

Subassembly 5: CAMERA FRAME:1

`CAMERA FRAME:1` under `root`

inspection view: occurrences at their CAD poses; not an assembly step; no sequence, fasteners or tool access implied.

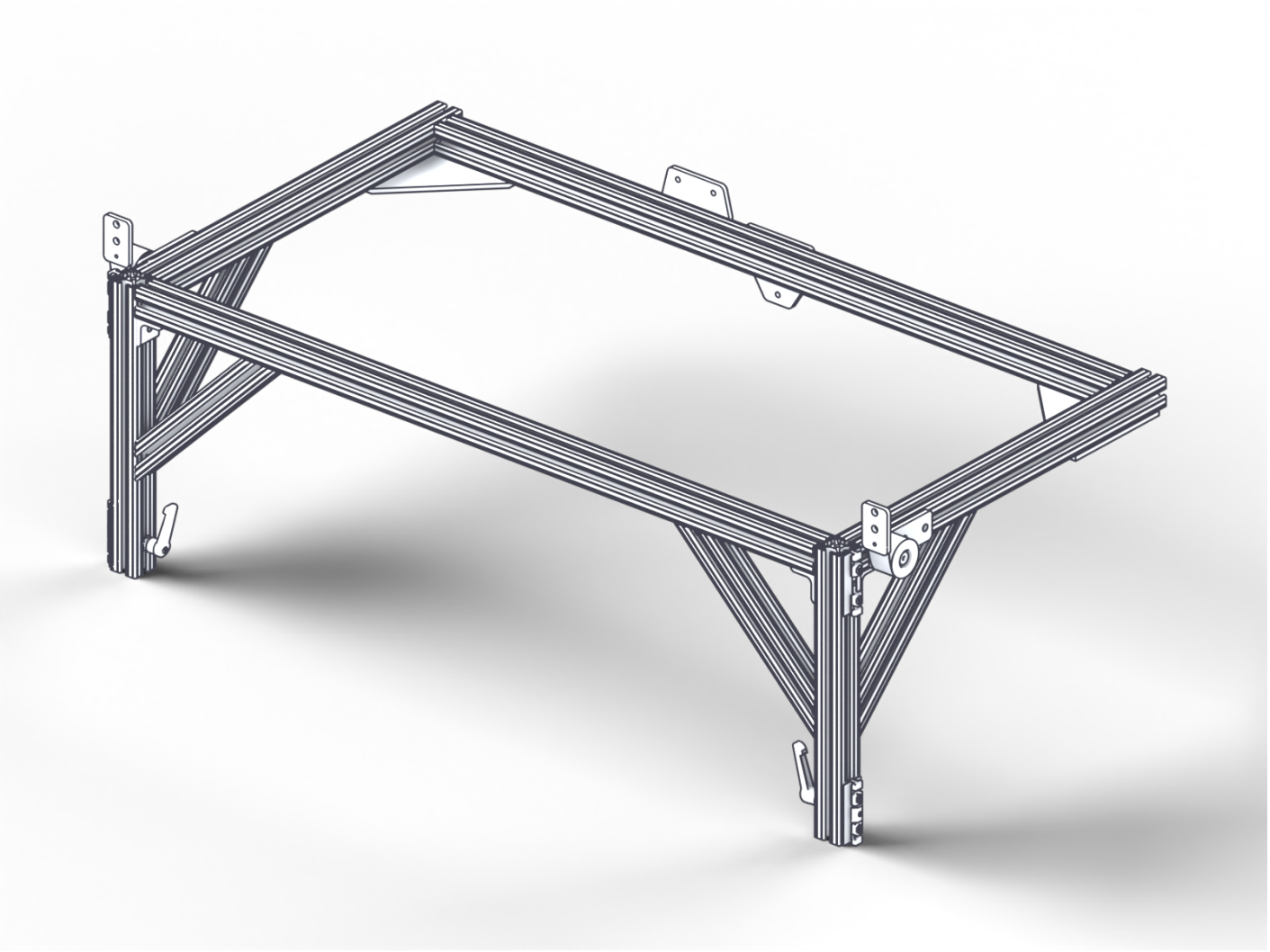
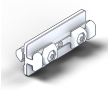
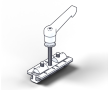
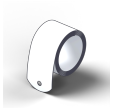
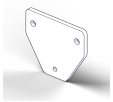


Figure 5: Subassembly 5: CAMERA FRAME:1, parts at their CAD positions (inspection view).

#	Picture	Part	In this subassembly
1		MCMASTER 5537T293 T-Slotted Framing	×2
2		MCMASTER 5537T429 T-Slotted Framing	×2
3		SLLB Enclosure 15in 47065T808 T-Slotted Framing	×2
4		SLLB Enclosure 22in 47065T808 T-Slotted Framing	×2
5		SLLB Enclosure 41in 47065T808 T-Slotted Framing	×2

#	Picture	Part	In this subassembly
6		MCMASTER 3136N149 T-Slotted Framing	×2
7		MCMASTER 47065T27 T-Slotted Framing	×4
8		MCMASTER 9293K121 Constant-Force Spring	×2
9		MCMASTER 47065T845 Silver Corner Bracket	×4
10		MCMASTER 5972K164 Ball Bearing	×4
11		MCMASTER 90265A164 18-8 Stainless Steel Shoulder Screw	×2
12		SPRING ROLLER	×2
13		ROLLER MOUNT PLATE	×2
14		SPRING MOUNT PLATE	×2
15		CAMERA MOUNT PLATE	×1
16		CAMERA MOUNT PLATE - TELEDYNE KINETIX	×1

36 occurrence(s); ids in manual.json.

Hardware to review at this joint (not an approved BOM)

- S8 Camera-frame connector and bracket hardware · candidate, requires review · required unknown (family, not specified); basis: 4 named 47065T845 brackets plus other connector/sliding parts · joint `camera_frame_joints`
- S12 Camera, spring and roller plate attachment hardware · candidate, requires review · required unknown (family, not specified); basis: 2 roller plates + 2 spring plates + 2 different camera plates · joint `camera_plates`
- S13 Spring end and roller/bearing retention hardware · candidate, requires review · required unknown (family, not specified); basis: 2 springs + 2 rollers + 4 bearings + 2 shoulder screws modeled · joint `spring_roller`
- S14 Camera-to-mount attachment · candidate, requires review · required unknown (family, not specified); basis: Camera plate variants present; actual camera selection unconfirmed · joint `camera_attachment`

Subassembly 6: FREE STANDING FRAME:1

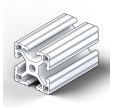
`FREE STANDING FRAME:1` under `root`

inspection view: occurrences at their CAD poses; not an assembly step; no sequence, fasteners or tool access implied.



Figure 6: Subassembly 6: FREE STANDING FRAME:1, parts at their CAD positions (inspection view).

#	Picture	Part	In this subassembly
24		SLLB Enclosure 44in 47065T808 T-Slotted Framing	×11
25		MCMASTER 47065T245	×8
26		SLLB Enclosure 96in 47065T808 T-Slotted Framing	×4
27		HOLKIE LEVELING CASTER	×4
28		CASTER MOUNTING PLATE	×4

#	Picture	Part	In this subassembly
29		SLLB Enclosure 2.5in 47065T808 T-Slotted Framing	×2
30		47065T271 ALUMINUM T-SLOTTED FRAMING EXTRUSION	×8
31		PANEL LOWER SUPPORT LEFT	×4
32		PANEL LOWER SUPPORT RIGHT	×4

49 occurrence(s); ids in manual.json.

Hardware to review at this joint (not an approved BOM)

- **S1** Caster-to-mounting-plate fastening · candidate, requires review · required unknown (family, not specified); basis: 4 caster assemblies; attachment points per caster unverified · joint `caster_to_plate`
- **S2** Caster-plate-to-frame fastening · candidate, requires review · required unknown (family, not specified); basis: 4 plates; frame attachments per plate unverified · joint `caster_plate_to_frame`
- **S6** Panel-lower-support fastening · candidate, requires review · required unknown (family, not specified); basis: 4 left + 4 right supports; points per support unverified · joint `lower_support_to_frame`
- **S7** Main-frame connector hardware and joint completeness · candidate, requires review · required unknown (family, not specified); basis: 11 long rails + 4 columns + 2 short rails; joint count requires review · joint `main_frame_joints`
- **S16** Removable-panel retention interface · candidate, requires review · required unknown (family, not specified); basis: 4 panels, 8 lower supports, 8 toggle clamps · joint `panel_retention`

Joint 1: caster_to_plate

context `FREE STANDING FRAME:1/CASTER ASSEMBLY:1` (faded, 0 other occurrence(s))

inspection view, not a step: the joint's occurrences at their CAD poses with the rest of the context subdued; no fastener, sequence or tool access implied.

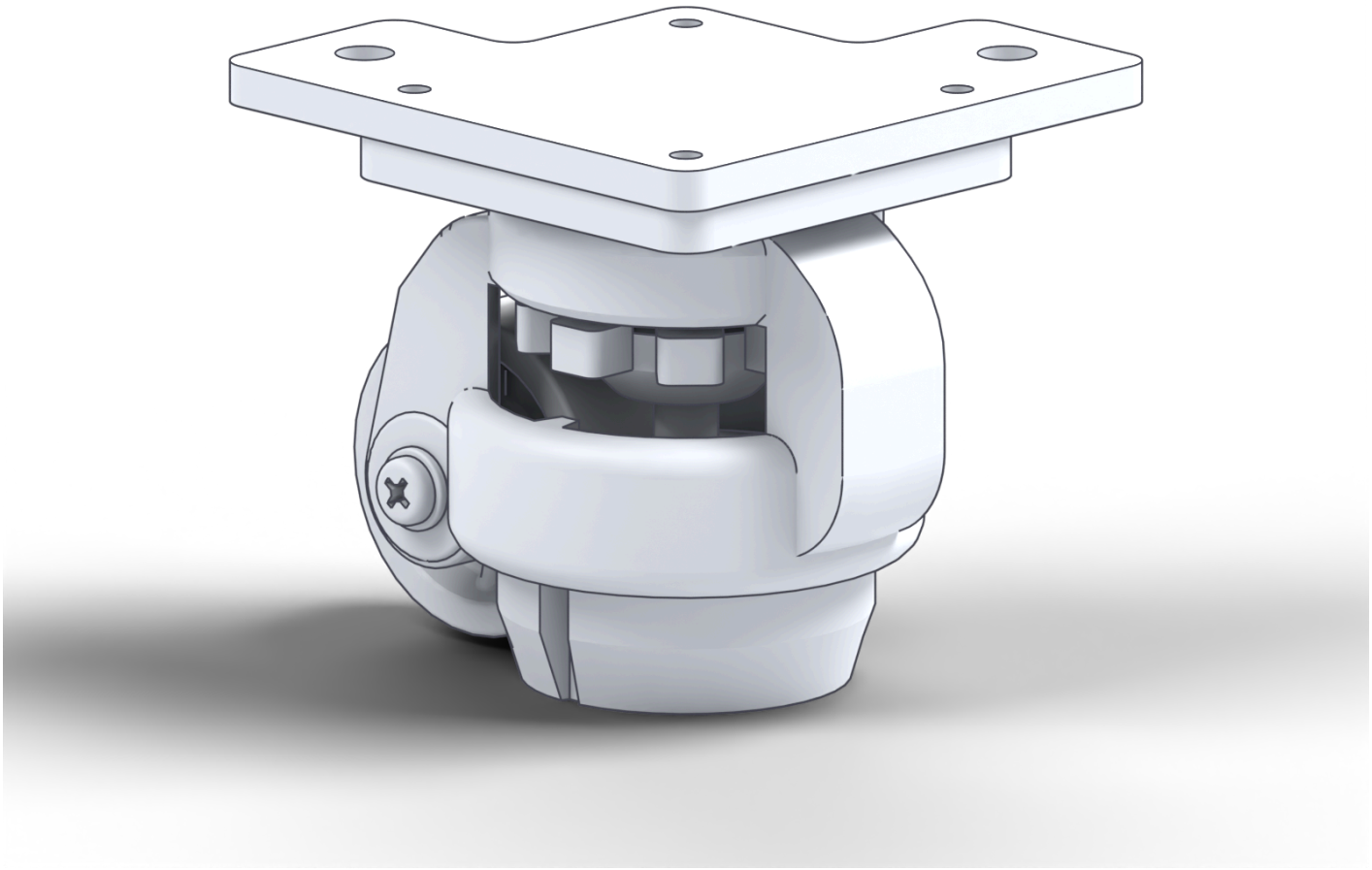
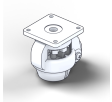


Figure 7: Joint 1: caster_to_plate, occurrences at their CAD positions (inspection view, not a step).

#	Picture	Occurrence in this joint
27		HOLKIE LEVELING CASTER `FREE STANDING FRAME:1/CASTER ASSEMBLY:1/HOLKIE LEVELING CASTER:1`
28		CASTER MOUNTING PLATE `FREE STANDING FRAME:1/CASTER ASSEMBLY:1/CASTER MOUNTING PLATE:1`

Hardware to review at this joint (not an approved BOM)

- S1 Caster-to-mounting-plate fastening · candidate, requires review · required unknown (family, not specified); basis: 4 caster assemblies; attachment points per caster unverified · joint `caster_to_plate`

Joint 2: caster_plate_to_frame

context `FREE STANDING FRAME:1` (faded, 48 other occurrence(s))

inspection view, not a step: the joint's occurrences at their CAD poses with the rest of the context subdued; no fastener, sequence or tool access implied.

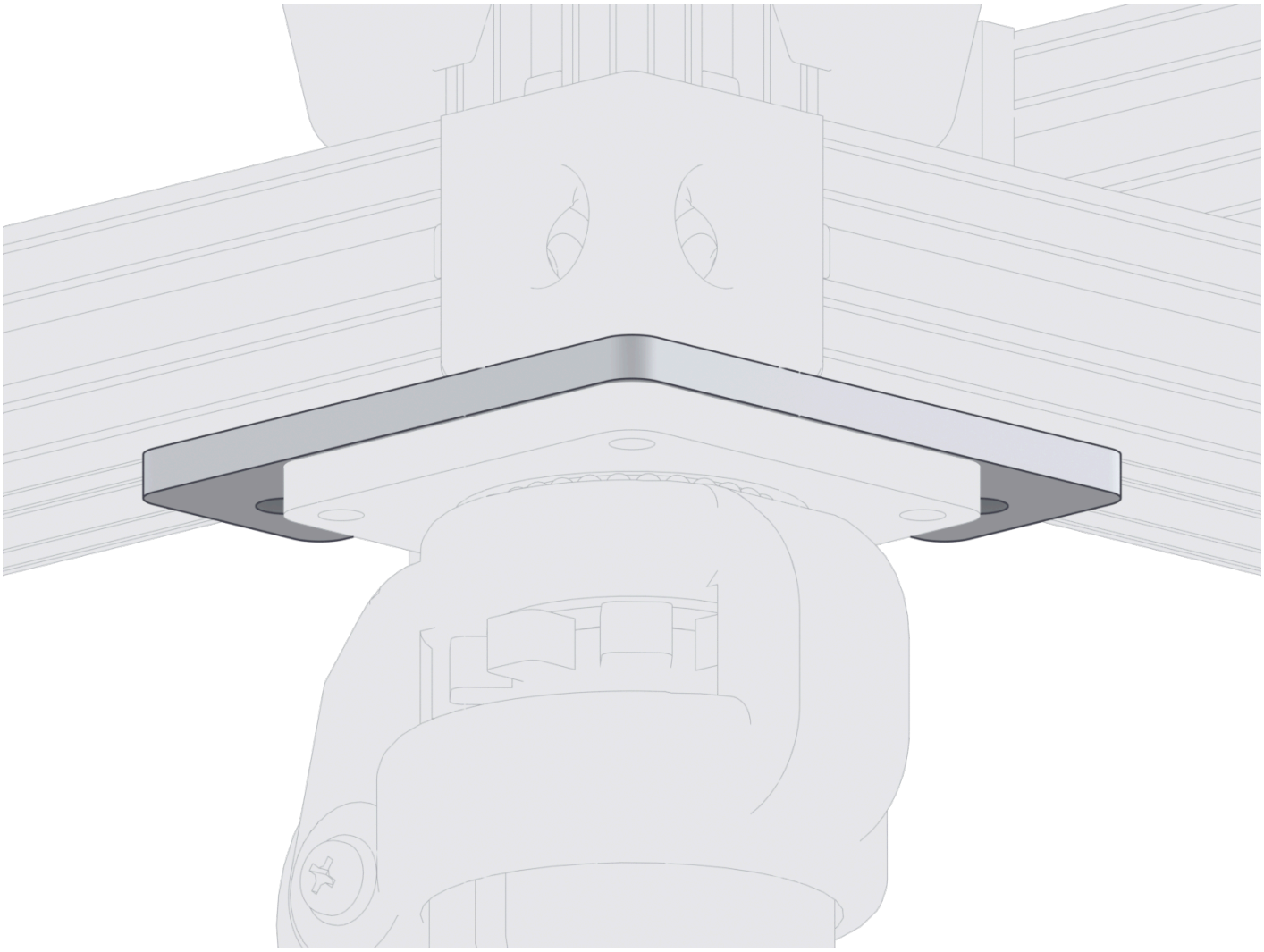


Figure 8: Joint 2: caster_plate_to_frame, occurrences at their CAD positions (inspection view, not a step).

#	Picture	Occurrence in this joint
28		CASTER MOUNTING PLATE `FREE STANDING FRAME:1/CASTER ASSEMBLY:1/CASTER MOUNTING PLATE:1`

Hardware to review at this joint (not an approved BOM)

- S2 Caster-plate-to-frame fastening · candidate, requires review · required unknown (family, not specified); basis: 4 plates; frame attachments per plate unverified · joint `caster_plate_to_frame`

Joint 3: handle_to_panel

context `PANEL ASSEMBLY:1` (faded, 2 other occurrence(s))

inspection view, not a step: the joint's occurrences at their CAD poses with the rest of the context subdued; no fastener, sequence or tool access implied.

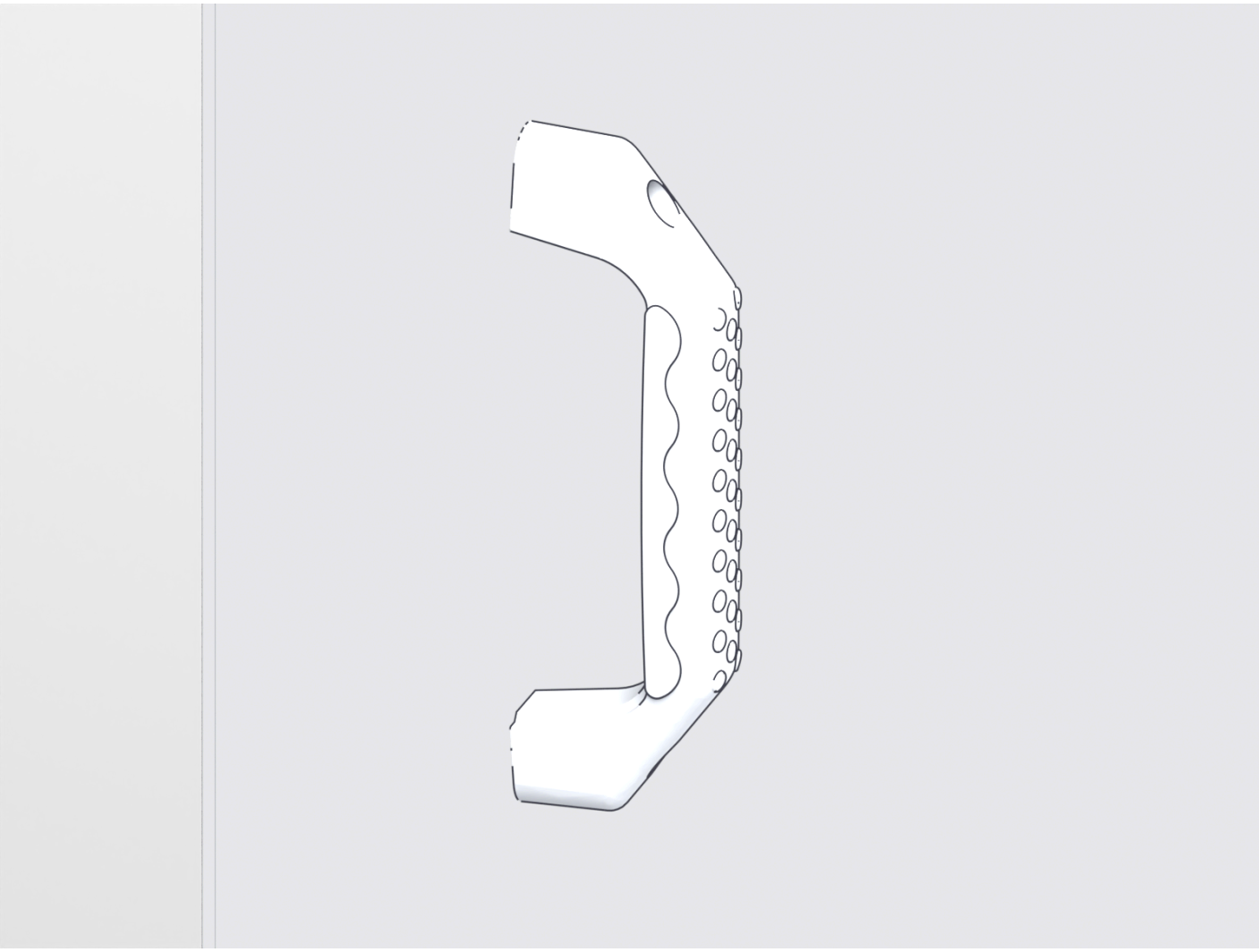


Figure 9: Joint 3: handle_to_panel, occurrences at their CAD positions (inspection view, not a step).

#	Picture	Occurrence in this joint
34		MCMASTER 1950A67 Unthreaded-Hole Pull Handle `PANEL ASSEMBLY:1/MCMASTER_1950A67_Unthreaded-Hole Pull Handle:1`

Hardware to review at this joint (not an approved BOM)

- S3 Pull-handle-to-panel fastening · candidate, requires review · required unknown (family, not specified); basis: 8 handles on 4 panels; fastening points per handle unverified · joint `handle_to_panel`

Joint 4: clamp_to_mount

context `TOGGLE CLAMP ASSEMBLY:1` (faded, 1 other occurrence(s))

inspection view, not a step: the joint's occurrences at their CAD poses with the rest of the context subdued; no fastener, sequence or tool access implied.

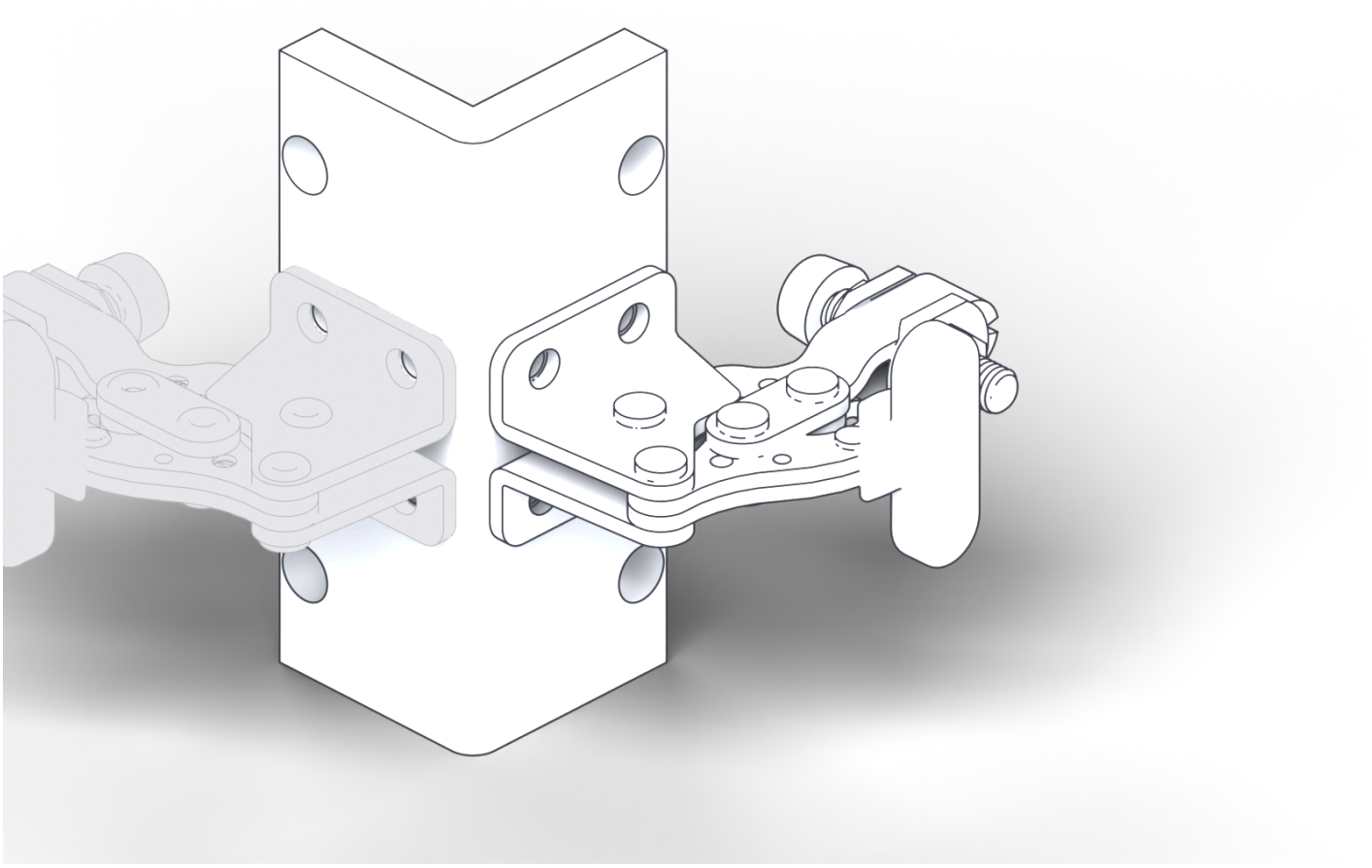
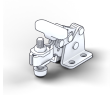


Figure 10: Joint 4: clamp_to_mount, occurrences at their CAD positions (inspection view, not a step).

#	Picture	Occurrence in this joint
36		MCMASTER 5004A11 Compact Hold-Down Toggle Clamp `TOGGLE CLAMP ASSEMBLY:1/MCMASTER 5004A11_Compact Hold-Down Toggle Clamp:1`
37		TOGGLE CLAMP MOUNT `TOGGLE CLAMP ASSEMBLY:1/TOGGLE CLAMP MOUNT:1`

Hardware to review at this joint (not an approved BOM)

- S4 Toggle-clamp-to-mount fastening · candidate, requires review · required unknown (family, not specified); basis: 8 clamps on 4 mounts; points per clamp unverified · joint `clamp_to_mount`

Joint 5: clamp_mount_to_frame

context `FREE STANDING FRAME:1` (faded, 49 other occurrence(s))

inspection view, not a step: the joint's occurrences at their CAD poses with the rest of the context subdued; no fastener, sequence or tool access implied.

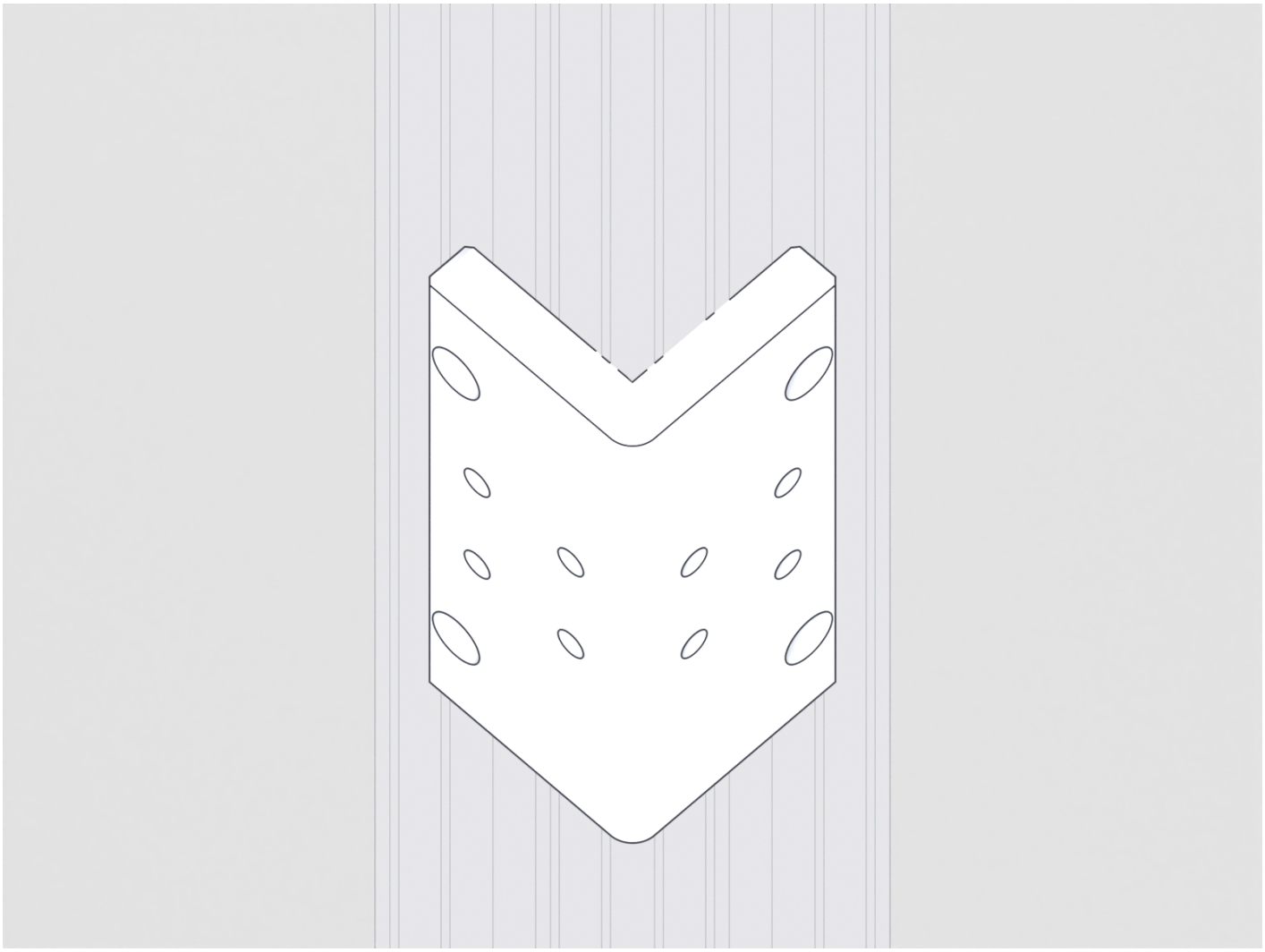
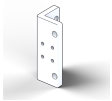


Figure 11: Joint 5: clamp_mount_to_frame, occurrences at their CAD positions (inspection view, not a step).

#	Picture	Occurrence in this joint
37		TOGGLE CLAMP MOUNT `TOGGLE CLAMP ASSEMBLY:1/TOGGLE CLAMP MOUNT:1`

Hardware to review at this joint (not an approved BOM)

- S5 Toggle-clamp-mount-to-frame fastening · candidate, requires review · required unknown (family, not specified); basis: 4 mounts; points per mount unverified · joint `clamp_mount_to_frame`

Joint 6: lower_support_to_frame

context `FREE STANDING FRAME:1` (faded, 48 other occurrence(s))

inspection view, not a step: the joint's occurrences at their CAD poses with the rest of the context subdued; no fastener, sequence or tool access implied.

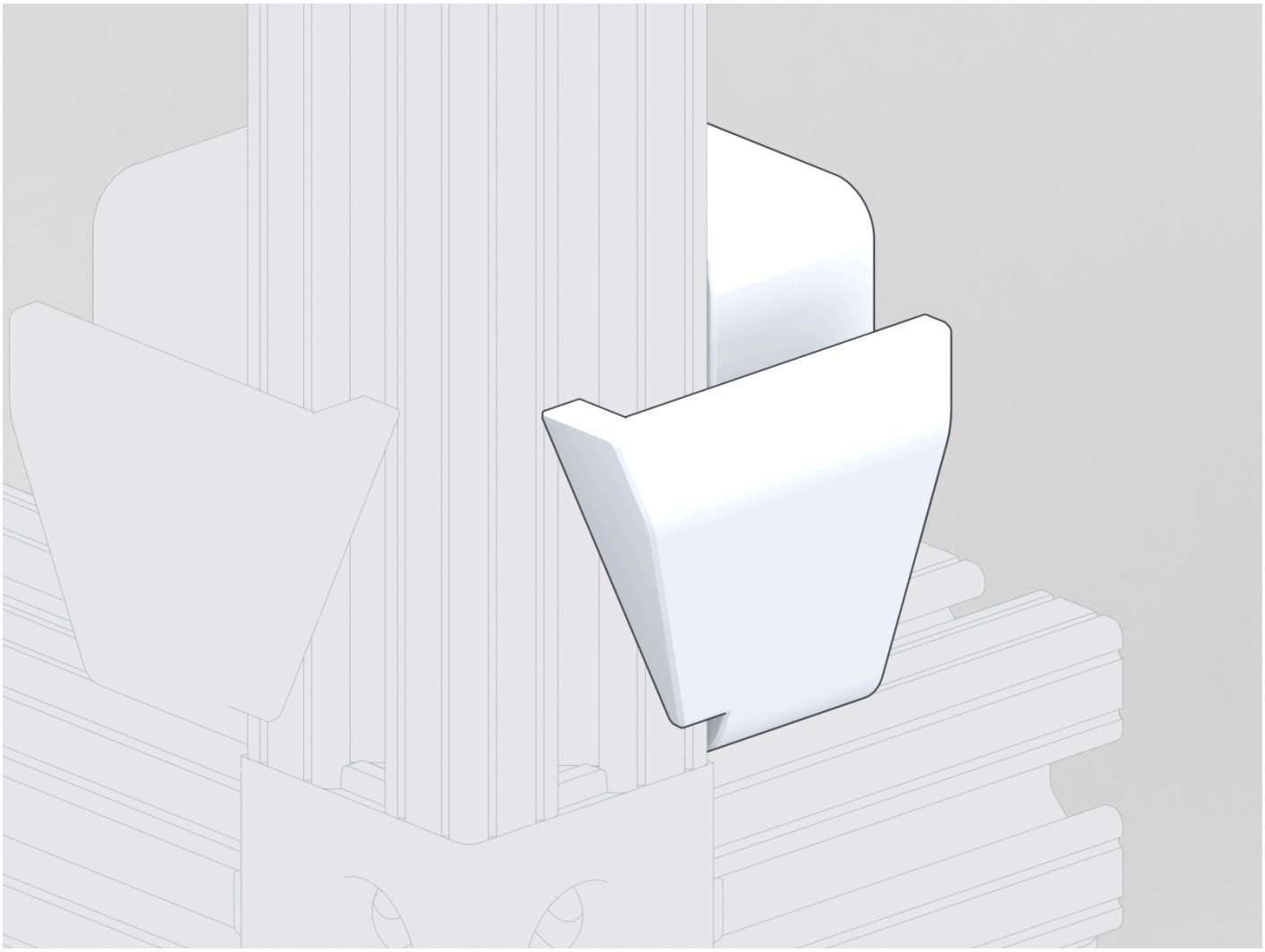


Figure 12: Joint 6: lower_support_to_frame, occurrences at their CAD positions (inspection view, not a step).

#	Picture	Occurrence in this joint
31		PANEL LOWER SUPPORT LEFT `FREE STANDING FRAME:1/PANEL LOWER SUPPORT LEFT:1`

Hardware to review at this joint (not an approved BOM)

- S6 Panel-lower-support fastening · candidate, requires review · required unknown (family, not specified); basis: 4 left + 4 right supports; points per support unverified · joint `lower_support_to_frame`

Joint 7: main_frame_joints

context `FREE STANDING FRAME:1` (faded, 48 other occurrence(s))

inspection view, not a step: the joint's occurrences at their CAD poses with the rest of the context subdued; no fastener, sequence or tool access implied.

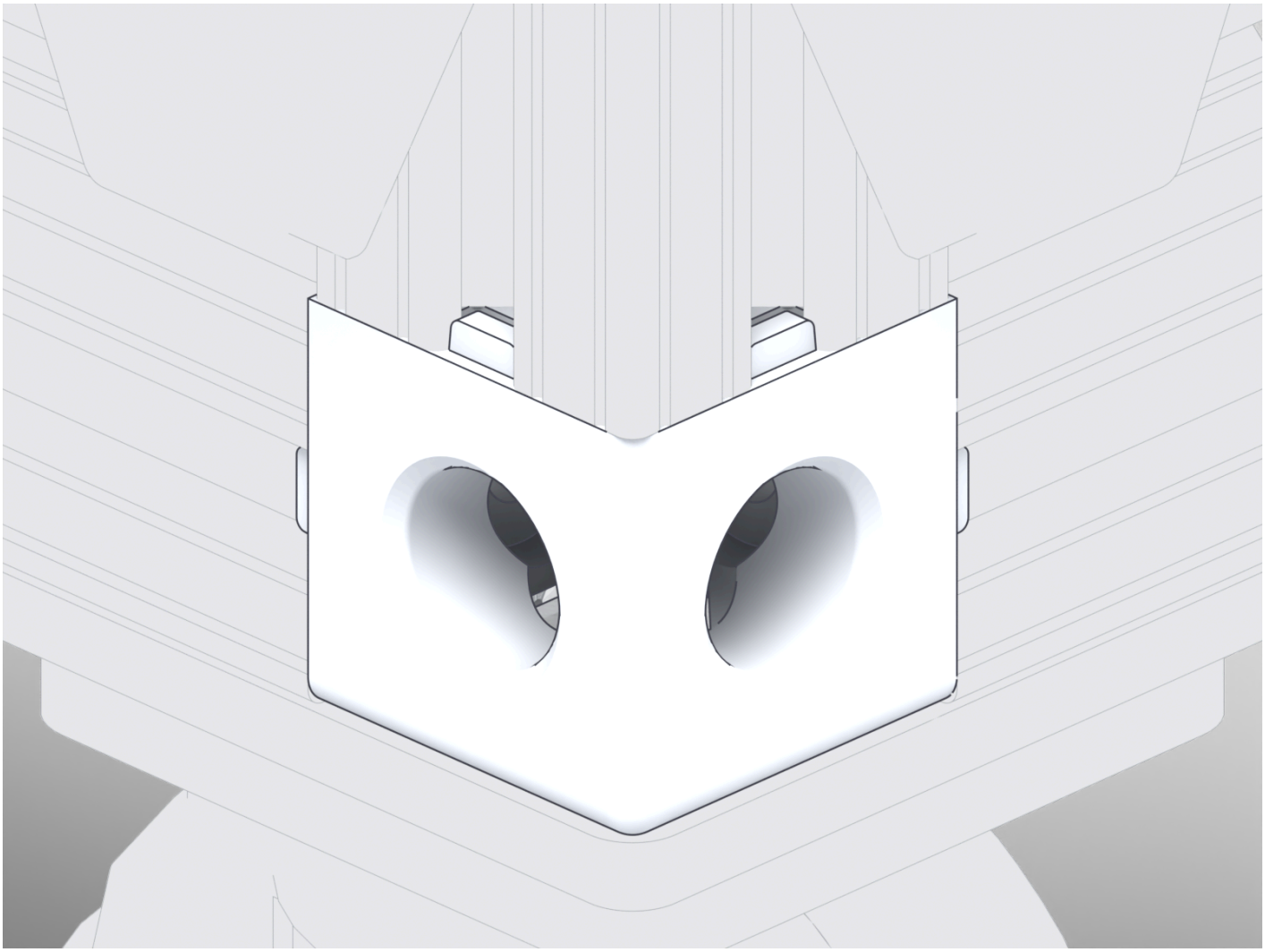


Figure 13: Joint 7: main_frame_joints, occurrences at their CAD positions (inspection view, not a step).

#	Picture	Occurrence in this joint
25		MCMASTER 47065T245 `FREE STANDING FRAME:1/MCMASTER 47065T245:1`

Hardware to review at this joint (not an approved BOM)

- S7 Main-frame connector hardware and joint completeness · candidate, requires review · required unknown (family, not specified); basis: 11 long rails + 4 columns + 2 short rails; joint count requires review · joint `main_frame_joints`

Joint 8: camera_frame_joints

context `CAMERA FRAME:1` (faded, 35 other occurrence(s))

inspection view, not a step: the joint's occurrences at their CAD poses with the rest of the context subdued; no fastener, sequence or tool access implied.

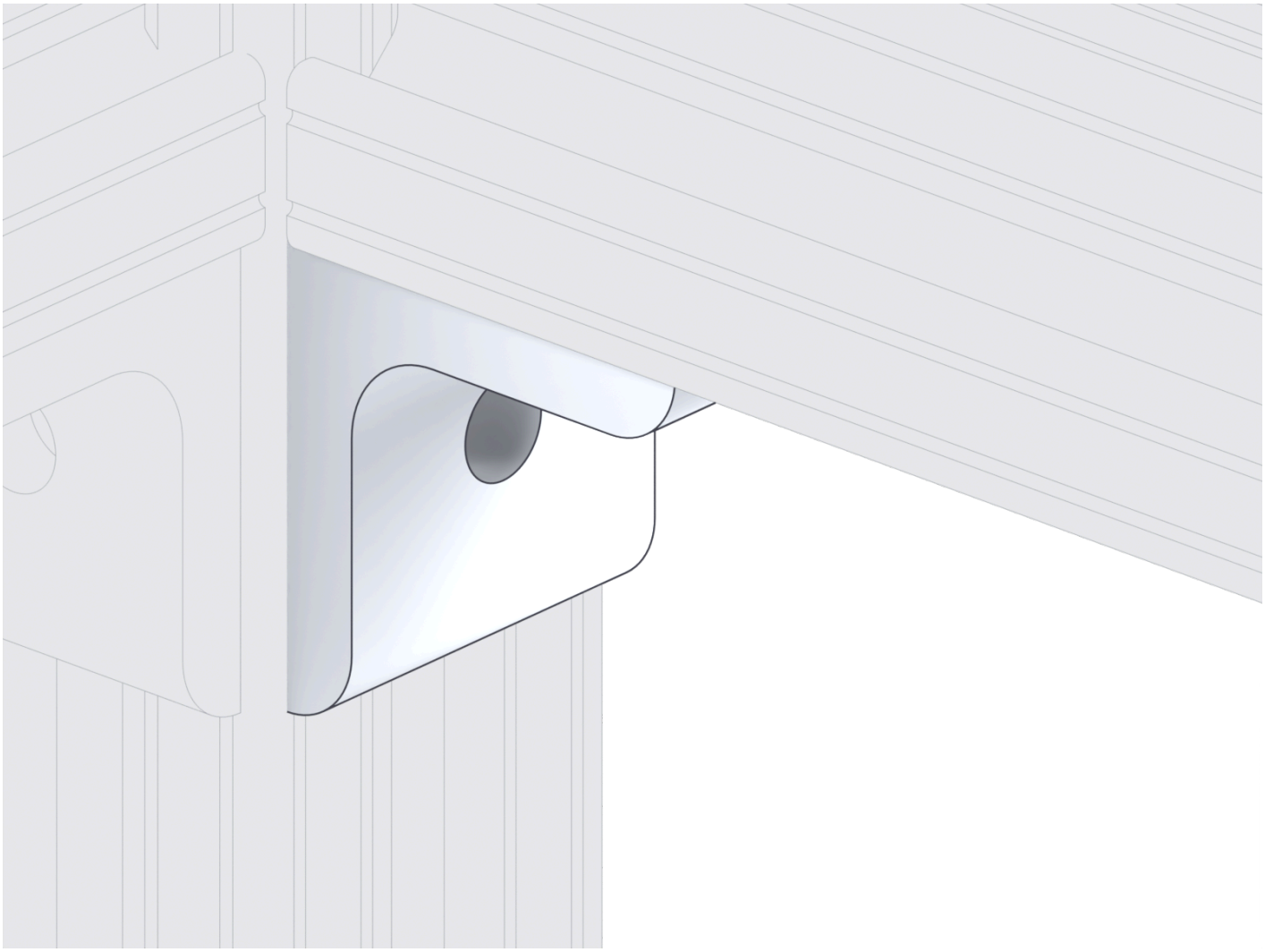


Figure 14: Joint 8: camera_frame_joints, occurrences at their CAD positions (inspection view, not a step).

#	Picture	Occurrence in this joint
9		MCMASTER 47065T845 Silver Corner Bracket `CAMERA FRAME:1/MCMASTER 47065T845_Silver Corner Bracket:1`

Hardware to review at this joint (not an approved BOM)

- S8 Camera-frame connector and bracket hardware · candidate, requires review · required unknown (family, not specified); basis: 4 named 47065T845 brackets plus other connector/sliding parts · joint `camera_frame_joints`

Joint 9: support_frame_corners

context `TANK SUPPORT FRAME:1` (faded, 21 other occurrence(s))

inspection view, not a step: the joint's occurrences at their CAD poses with the rest of the context subdued; no fastener, sequence or tool access implied.

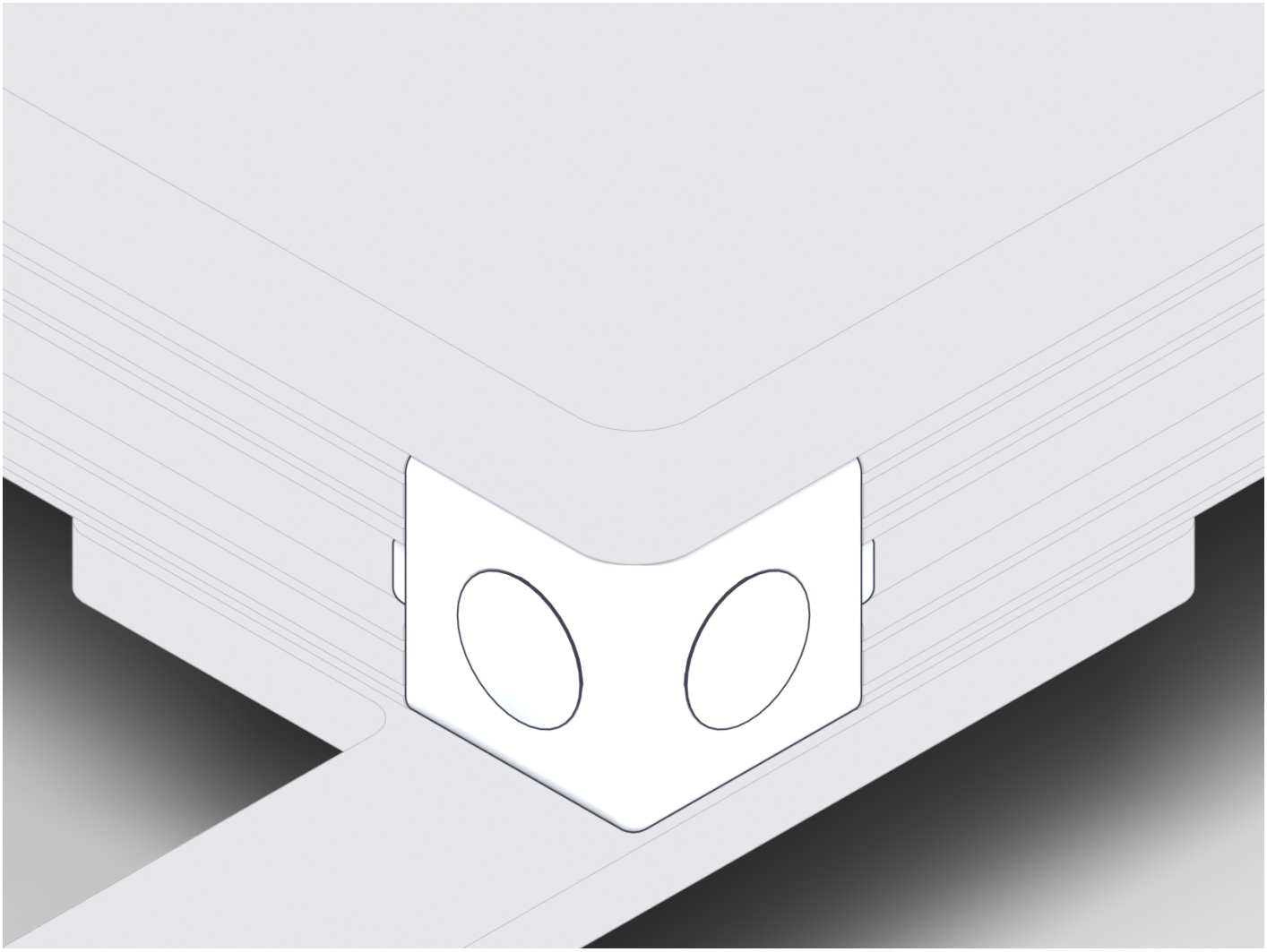
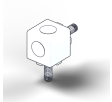


Figure 15: Joint 9: support_frame_corners, occurrences at their CAD positions (inspection view, not a step).

#	Picture	Occurrence in this joint
18		T SLOT CORNER 47065T242 `TANK SUPPORT FRAME:1/T SLOT CORNER 47065T242:1`

Hardware to review at this joint (not an approved BOM)

- S9 Tank-support-frame corner hardware · candidate, requires review · required unknown (family, not specified); basis: 8 modeled corner connectors across 2 support frames · joint `support_frame_corners`

Joint 10: support_brackets

context `TANK SUPPORT FRAME:1` (faded, 21 other occurrence(s))

inspection view, not a step: the joint's occurrences at their CAD poses with the rest of the context subdued; no fastener, sequence or tool access implied.

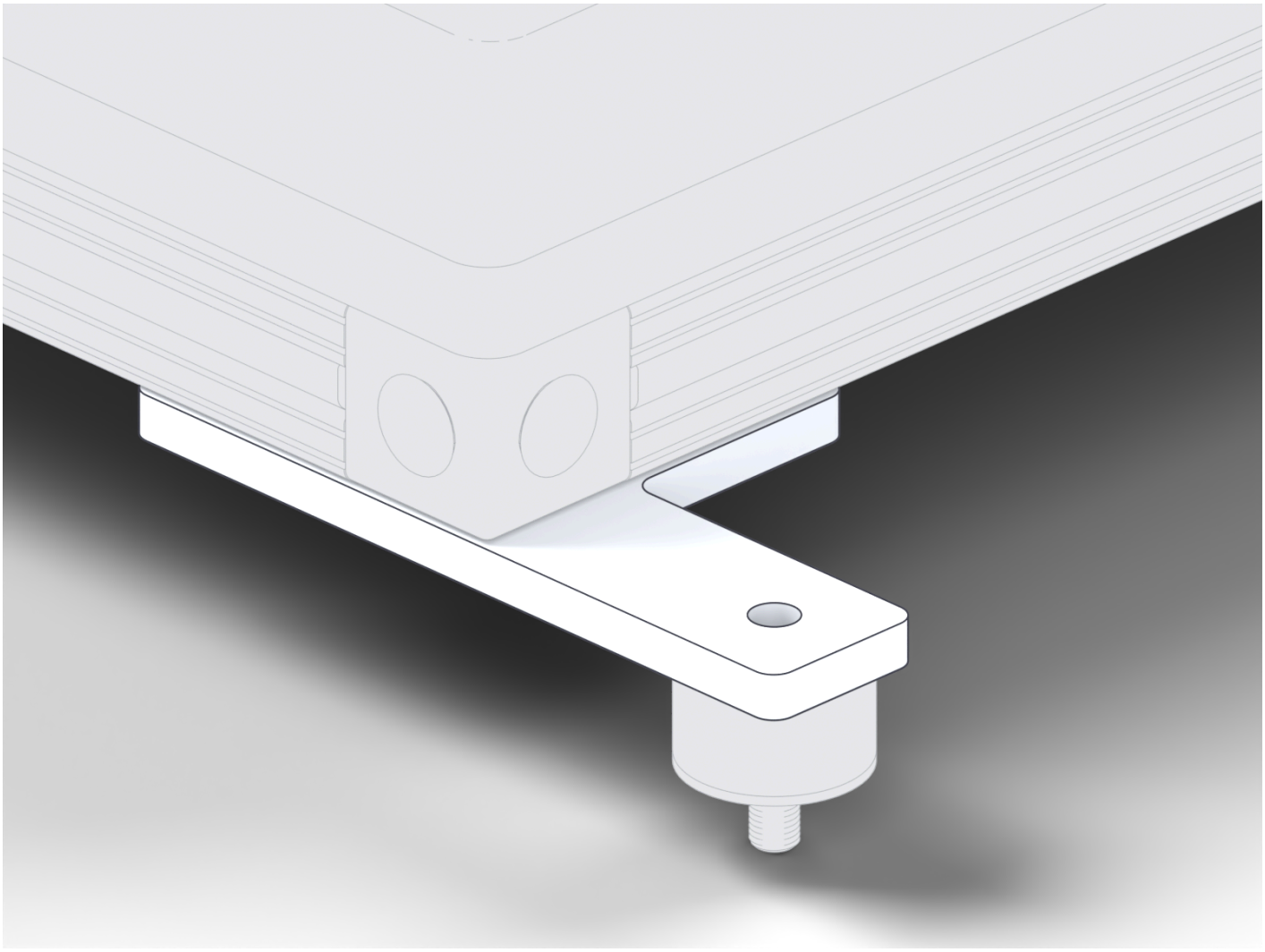


Figure 16: Joint 10: support_brackets, occurrences at their CAD positions (inspection view, not a step).

#	Picture	Occurrence in this joint
23		TANK SUPPORT BRACKET `TANK SUPPORT FRAME:1/TANK SUPPORT BRACKET:1`

Hardware to review at this joint (not an approved BOM)

- S10 Tank-support-bracket attachment hardware · candidate, requires review · required unknown (family, not specified); basis: 8 brackets across 2 support frames · joint `support_brackets`

Joint 11: vibration_mounts

context `TANK SUPPORT FRAME:1` (faded, 20 other occurrence(s))

inspection view, not a step: the joint's occurrences at their CAD poses with the rest of the context subdued; no fastener, sequence or tool access implied.

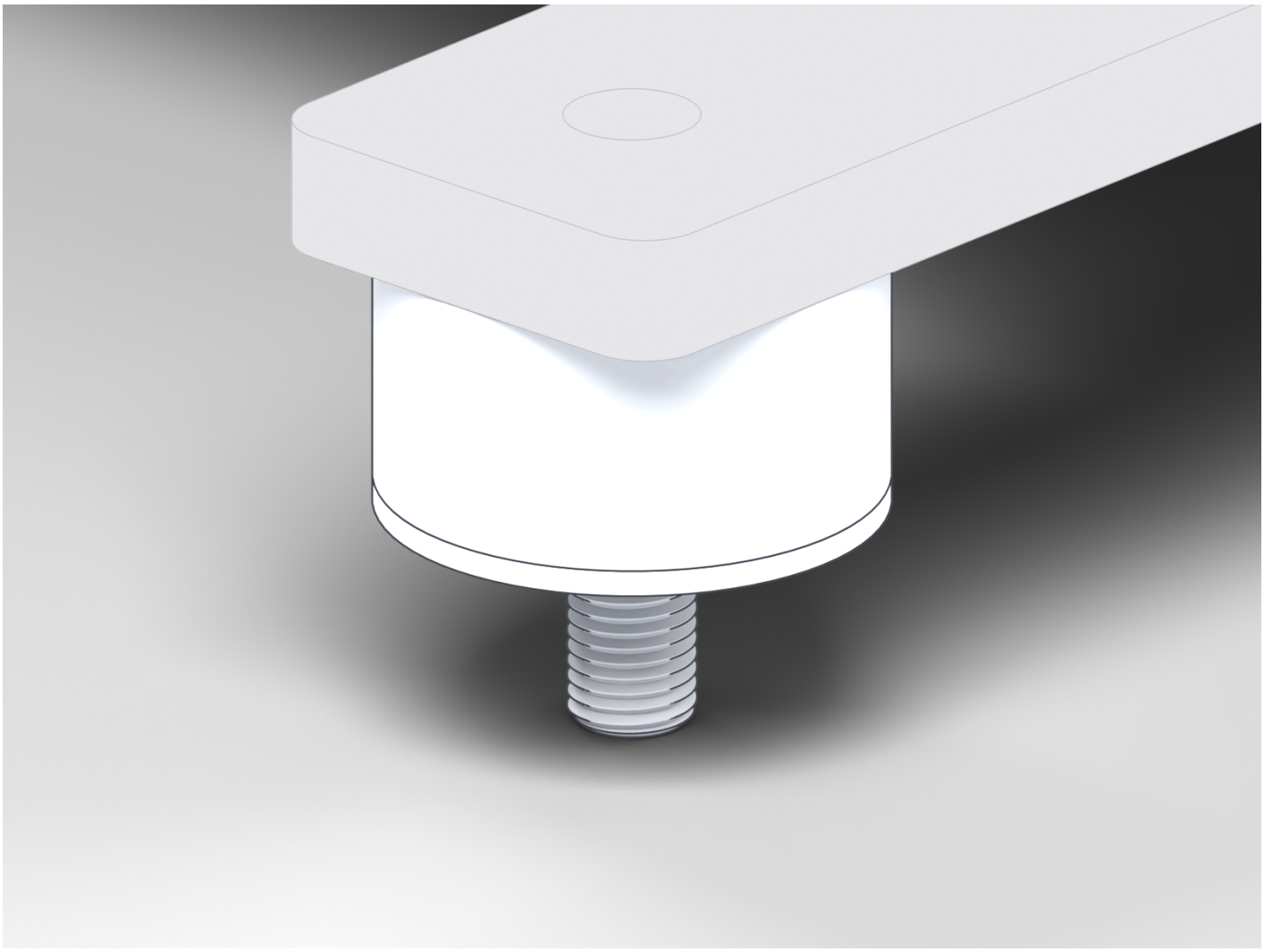
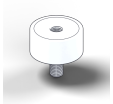


Figure 17: Joint 11: vibration_mounts, occurrences at their CAD positions (inspection view, not a step).

#	Picture	Occurrence in this joint
19		MCMASTER 9378K24 Natural Rubber Vibration-Damping Sandwich Mount `TANK SUPPORT FRAME:1/MCMASTER 9378K24_Natural Rubber Vibration-Damping Sandwich Mount:1`
20		MCMASTER 90313A107 18-8 Stainless Steel Oversized Washer `TANK SUPPORT FRAME:1/MCMASTER 90313A107_18-8 Stainless Steel Oversized Washer:4`

Hardware to review at this joint (not an approved BOM)

- S11 Vibration-mount retention hardware · candidate, requires review · required unknown (family, not specified); basis: 8 mounts + 8 oversized washers modeled · joint `vibration_mounts`

Joint 12: camera_plates

context `CAMERA FRAME:1` (faded, 34 other occurrence(s))

inspection view, not a step: the joint's occurrences at their CAD poses with the rest of the context subdued; no fastener, sequence or tool access implied.

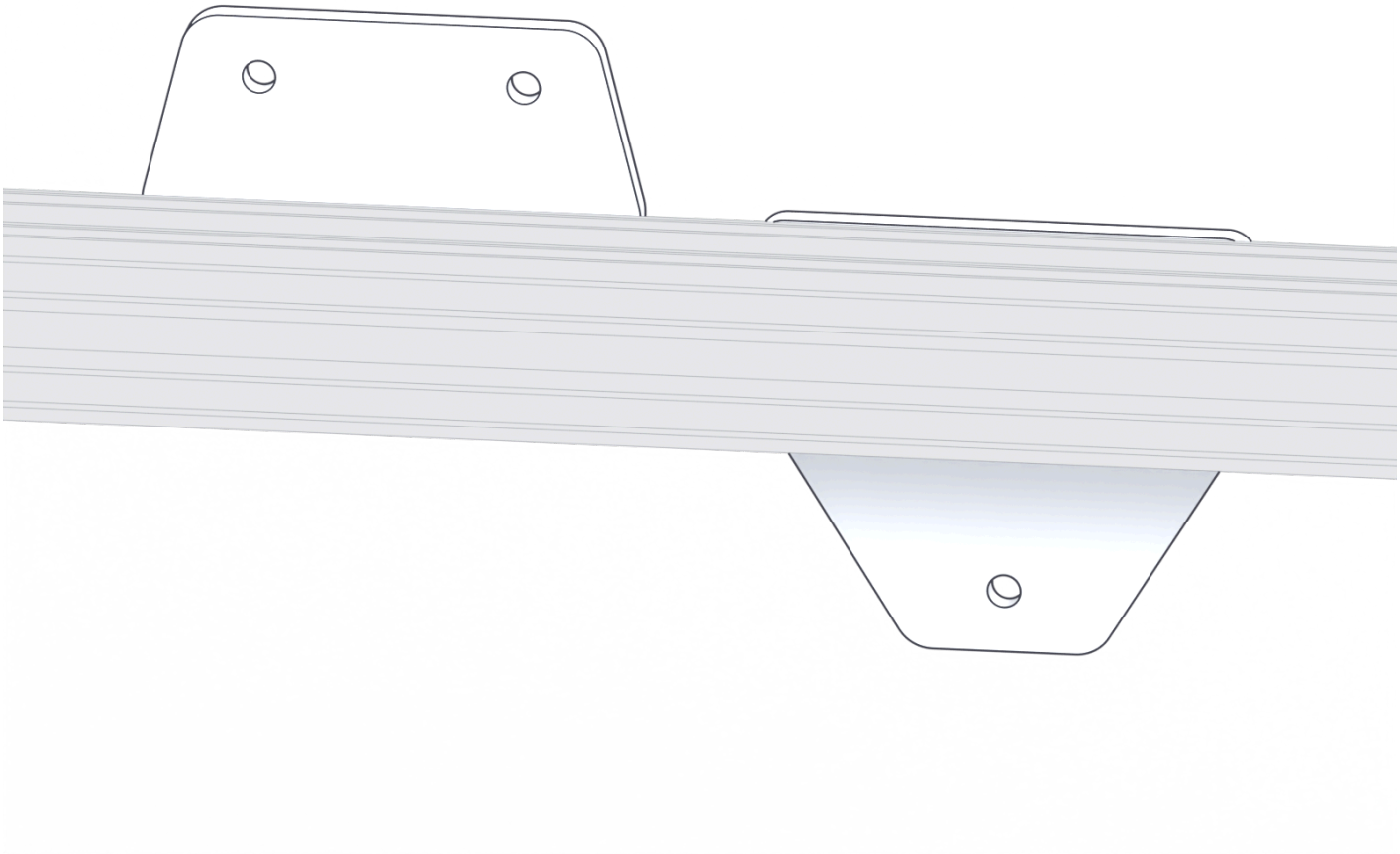
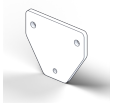
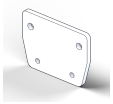


Figure 18: Joint 12: camera_plates, occurrences at their CAD positions (inspection view, not a step).

#	Picture	Occurrence in this joint
15		CAMERA MOUNT PLATE `CAMERA FRAME:1/CAMERA MOUNT PLATE:1`
16		CAMERA MOUNT PLATE - TELEDYNE KINETIX `CAMERA FRAME:1/CAMERA MOUNT PLATE - TELEDYNE KINETIX:1`

Hardware to review at this joint (not an approved BOM)

- S12 Camera, spring and roller plate attachment hardware · candidate, requires review · required unknown (family, not specified); basis: 2 roller plates + 2 spring plates + 2 different camera plates · joint `camera_plates`

Joint 13: spring_roller

context `CAMERA FRAME:1` (faded, 31 other occurrence(s))

inspection view, not a step: the joint's occurrences at their CAD poses with the rest of the context subdued; no fastener, sequence or tool access implied.

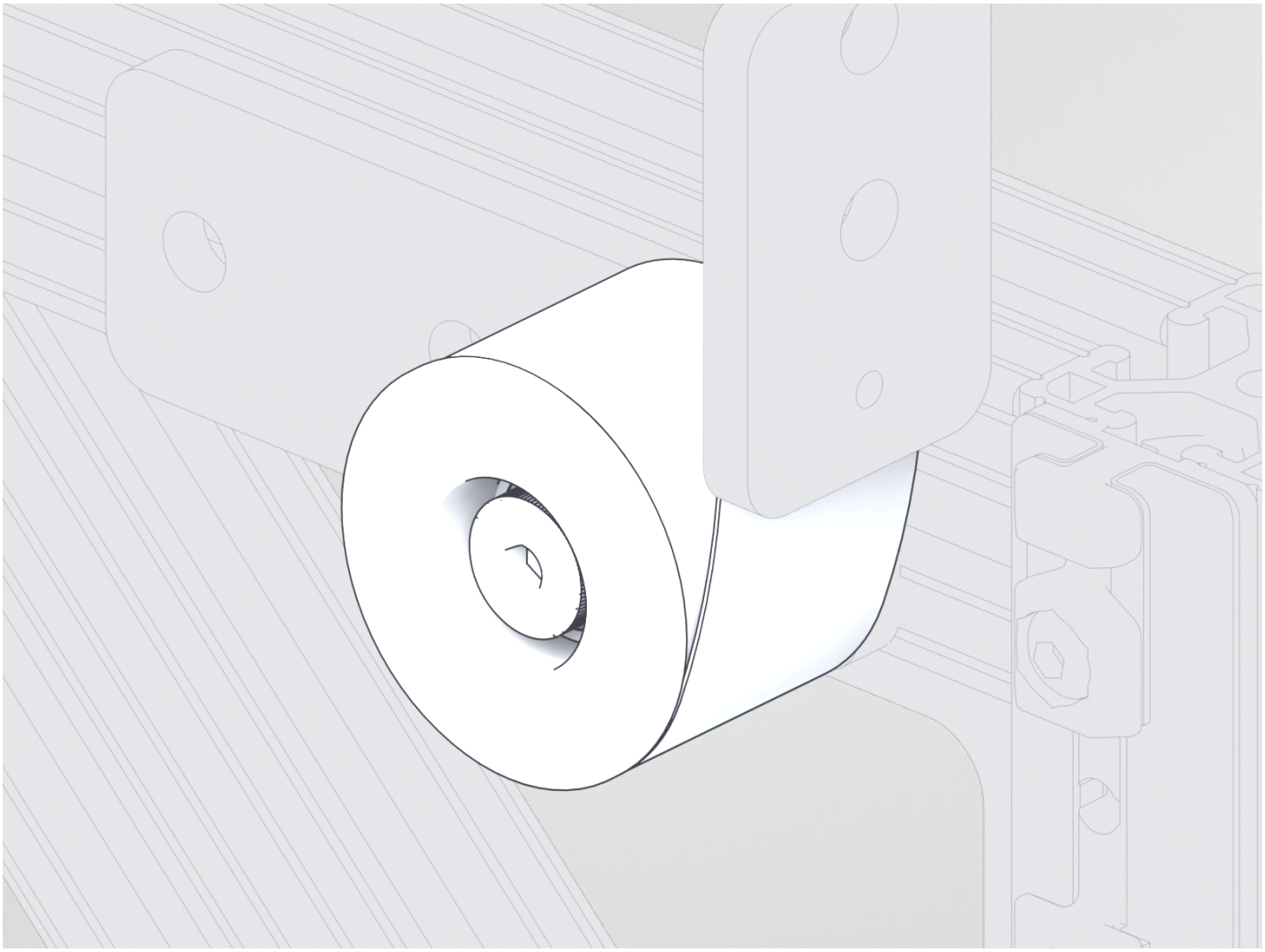
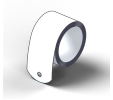


Figure 19: Joint 13: spring_roller, occurrences at their CAD positions (inspection view, not a step).

#	Picture	Occurrence in this joint
12		SPRING ROLLER `CAMERA FRAME:1/SPRING ROLLER:1`
10		MCMASTER 5972K164 Ball Bearing `CAMERA FRAME:1/MCMASTER 5972K164_Ball Bearing:2`
11		MCMASTER 90265A164 18-8 Stainless Steel Shoulder Screw `CAMERA FRAME:1/MCMASTER 90265A164_18-8 Stainless Steel Shoulder Screw:1`
8		MCMASTER 9293K121 Constant-Force Spring `CAMERA FRAME:1/MCMASTER 9293K121_Constant-Force Spring:1`
10		MCMASTER 5972K164 Ball Bearing `CAMERA FRAME:1/MCMASTER 5972K164_Ball Bearing:1`

Hardware to review at this joint (not an approved BOM)

- S13 Spring end and roller/bearing retention hardware · candidate, requires review · required unknown (family, not specified); basis: 2 springs + 2 rollers + 4 bearings + 2 shoulder screws modeled · joint `spring_roller`

Joint 14: camera_attachment

context `CAMERA FRAME:1` (faded, 34 other occurrence(s))

inspection view, not a step: the joint's occurrences at their CAD poses with the rest of the context subdued; no fastener, sequence or tool access implied.

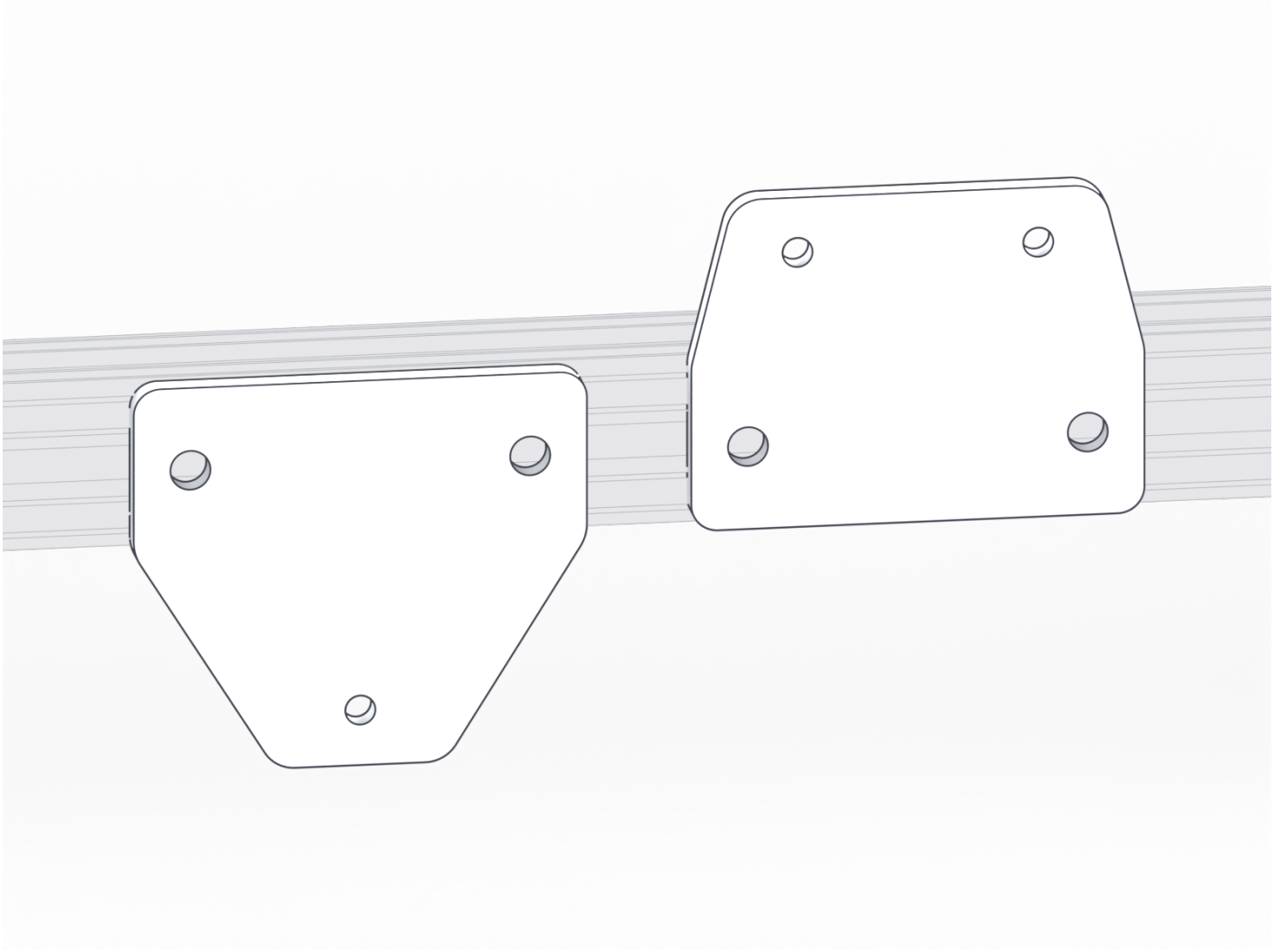
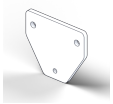
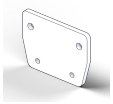


Figure 20: Joint 14: camera_attachment, occurrences at their CAD positions (inspection view, not a step).

#	Picture	Occurrence in this joint
15		CAMERA MOUNT PLATE `CAMERA FRAME:1/CAMERA MOUNT PLATE:1`
16		CAMERA MOUNT PLATE - TELEDYNE KINETIX `CAMERA FRAME:1/CAMERA MOUNT PLATE - TELEDYNE KINETIX:1`

Hardware to review at this joint (not an approved BOM)

- S14 Camera-to-mount attachment · candidate, requires review · required unknown (family, not specified); basis: Camera plate variants present; actual camera selection unconfirmed · joint `camera_attachment`

Joint 15: top_panel

context `root` (faded, 154 other occurrence(s))

inspection view, not a step: the joint's occurrences at their CAD poses with the rest of the context subdued; no fastener, sequence or tool access implied.



Figure 21: Joint 15: top_panel, occurrences at their CAD positions (inspection view, not a step).

#	Picture	Occurrence in this joint
35		ENCLOSURE TOP PANEL `ENCLOSURE TOP PANEL:1`

Hardware to review at this joint (not an approved BOM)

- S15 Top-panel retention and edge/support interface · candidate, requires review · required unknown (family, not specified); basis: 1 enclosure top panel · joint `top_panel`

Joint 16: panel_retention

context `FREE STANDING FRAME:1` (faded, 47 other occurrence(s))

inspection view, not a step: the joint's occurrences at their CAD poses with the rest of the context subdued; no fastener, sequence or tool access implied.

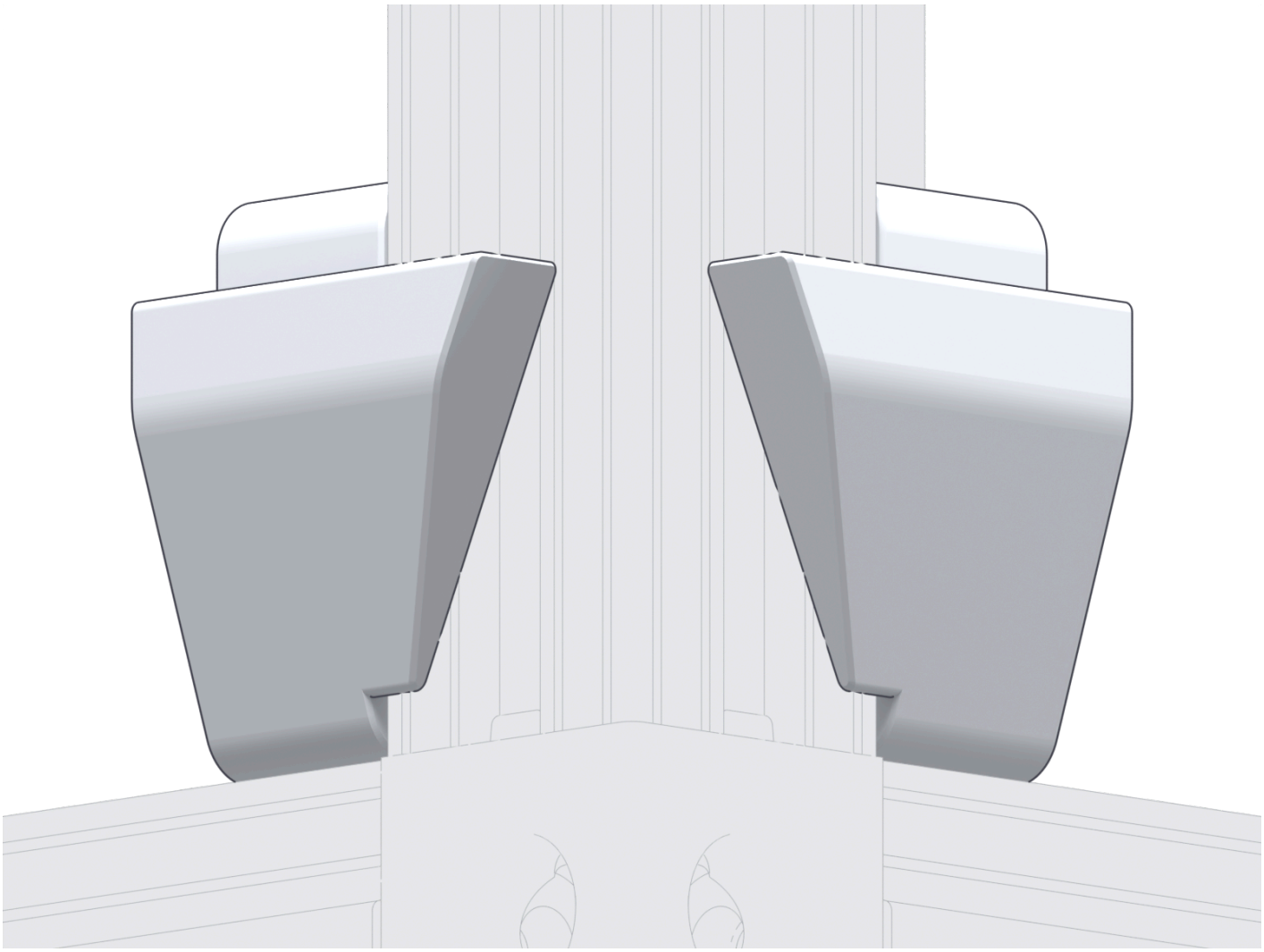


Figure 22: Joint 16: panel_retention, occurrences at their CAD positions (inspection view, not a step).

#	Picture	Occurrence in this joint
31		PANEL LOWER SUPPORT LEFT `FREE STANDING FRAME:1/PANEL LOWER SUPPORT LEFT:1`
32		PANEL LOWER SUPPORT RIGHT `FREE STANDING FRAME:1/PANEL LOWER SUPPORT RIGHT:2`

Hardware to review at this joint (not an approved BOM)

- S16 Removable-panel retention interface · candidate, requires review · required unknown (family, not specified); basis: 4 panels, 8 lower supports, 8 toggle clamps · joint `panel_retention`

Finished assembly



Figure 23: All parts at their CAD positions.

Steps

No assembly sequence has been authored for this scene.

Build record

generated	2026-10-22T18:37:42Z
scene	free-standing-enclosure-991056: Free Standing Enclosure (Vault file-version 991056) native snapshot (37 parts, 155 occurrences, 17 assemblies)
source kind	autodesk-inventor-vault-snapshot
source root_version_id	991056
source exporter	windows/Export-PartAssets.ps1 on Inventor 2025.3
source qualification	geometry/hierarchy snapshot; not a reviewed assembly procedure or stock inventory
source occurrence_id_scope	snapshot-scoped occurrence paths (Inventor occurrence names); stable only within this snapshot
scene.json	`ac203e5656abdeeadbadf5f047e31f244e01f8938001c5747a44808ffc4ed070`
inventory.json	`0eed5897bb238c1334e92e19927db8573480f3c7d4778099fa02b09e605298de`
supplement.json	`c9950ec3f72db98f8d3563adaec0fb48ba50fce689fbb905dbe050ce0e4081ba`
assets	37 STEP files; hashes in manual.json
FreeCAD	1.1.3; deflection 0.05 mm / 10.0°, budget 50000 triangles per part; meshes.json `91c004502b7a089aac81b6d07810e5eae461b3174bfaaca41c468ba3a5a6f414`
Blender	4.5.14 LTS, engine CYCLES, denoiser OPENIMAGEDENOISE, samples 64, long side 1600 px; line art on up to 600000 visible triangles; settings `de5362725e51f850f9ae2abe69e598ccfc9de12271f91db3d29b9426bc8bb3bb`
reused views	66 of 67 images copied from an earlier run with the same inputs, meshes and engine (build/enclosure-snapshot-991056/renders7/render-receipt.json); per-view record in manual.json
images	67 renders; hashes in manual.json
supplement.json	20 row(s): 16 existing design, 4 new design; never added to CAD quantities
generator	python 3.12.12; platform Linux-6.12.113+deb13-amd64-x86_64-with-glibc2.41; package mechanical-docs

Review state draft; physical assembly not validated.